

Recent application of spectroscopy for the detection of microalgae life information: A review

Jing-Yan Liu, Li-Hua Zeng & Zhen-Hui Ren

To cite this article: Jing-Yan Liu, Li-Hua Zeng & Zhen-Hui Ren (2020) Recent application of spectroscopy for the detection of microalgae life information: A review, Applied Spectroscopy Reviews, 55:1, 26-59, DOI: [10.1080/05704928.2018.1509345](https://doi.org/10.1080/05704928.2018.1509345)

To link to this article: <https://doi.org/10.1080/05704928.2018.1509345>



Published online: 15 Jan 2019.



Submit your article to this journal [↗](#)



Article views: 1501



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 27 View citing articles [↗](#)



Recent application of spectroscopy for the detection of microalgae life information: A review

Jing-Yan Liu*, Li-Hua Zeng*, and Zhen-Hui Ren

College of Mechanical and Electrical Engineering, Hebei Agricultural University, Baoding, China

ABSTRACT

Monitoring the growth status of microalgae is very important. Traditional techniques for the detection and evaluation of life information during the growth of microalgae are tedious, laborious, destructive and time-consuming. Objective quality assessments and efficacious safety surveillance for microalgal growth information are inseparable from innovative techniques. This review focuses on the applications of several valuable spectroscopic techniques, including visible and near-infrared spectroscopy, Raman spectroscopy, Fourier Transform Infrared (FTIR) spectroscopy and fluorescence spectroscopy, for the rapid and nondestructive detection of microalgae growth information, such as pigment content, oil content, biomass content and identification of algae species. The fundamentals of Spectroscopic Techniques are presented and discussed. Then, we illustrate the most typically and commonly used chemometric analysis, such as spectral preprocessing and modeling methods. Moreover, the advantages and disadvantages of these technologies are discussed, and the applications of these technologies are summarized.

KEYWORDS

Microalgae; visible and near-infrared spectroscopy; Raman spectroscopy; Fourier Transform Infrared (FTIR) spectroscopy; fluorescence spectroscopy

Introduction

There are many species of microalgae that are widely distributed on the earth. Currently, there are more than 20,000 species of microalgae globally known, accounting for 70% of all algae species (1). Compared with traditional field crops, microalgae have the advantage of fast growth, high photosynthetic efficiency, short doubling time and high yield per unit of area. Therefore, there are broad prospects for the application and development of microalgae in energy, food industry, biotechnology, pharmaceutical industry, animal feed, environmental monitoring, pollution control and other fields. Microalgae have an important influence on the ecological environment and biology and have attracted worldwide attention (2, 3).

Microalgae are rich in nutrients and physiologically active ingredients, such as protein, Omega-3 polyunsaturated fatty acids, polysaccharides, carotene, various vitamins, glycerin and trace elements (4–6). The content and composition of nutrients in microalgae generally depend on the species of algae and the culture environment. The best harvest time for different microalgae in different growth environments is dynamic.

CONTACT Li-Hua Zeng ✉ Zenglh@cau.edu.cn; Zhen-hui Ren ✉ Renzh@hebau.edu.cn College of Mechanical and Electrical Engineering, Hebei Agricultural University, No. 289 Lingyusi Street, Baoding, Hebei 071001, China.

*Equally contributed to the work.

The environmental parameters that affect microalgae growth are the most important factors that directly relate to the growth of microalgae (7). Therefore, the rapid acquisition of microalgae life information can be used to effectively optimize microalgae growth environment parameters to improve production efficiency, which has been a concern of researchers for years (8).

Currently, the conventional method for obtaining microalgae growth information is mainly via manual observation or wet chemical methods (8). For example, a researcher can evaluate the growth status of the microalgae by observing the color and turbidity of the microalgae culture medium. However, the quantitative detection of nutrients is usually carried out by experienced evaluators using chemical analysis methods (e.g., the gravity method, Lowry method and Kjeldahl method) (9). Traditional methods, which are based on the rich experience of the surveyor, are not suitable for the large and continuous measurement of microalgae due to the high professional requirements and time-consuming. In addition, the extraction of samples easily causes microalgae pollution, which affects production efficiency, and the detection process is easily influenced by human emotions and other factors that result in large deviations. Therefore, it is necessary to acquire and control digital information regarding the microalgae production process through modern information technology (8). Optimizing the biological system to control biological processes is an important way to improve the efficiency of biological cultivation, reduce costs and increase economic benefits (9). Optimizing the biological system is also an important direction for the construction of information technology for the microalgae biomass industry (8).

Many techniques have been applied to obtain microalgae growth information, such as the pigment method (10, 11), flow cytometry (12), fluorescence probe technology (13), high-performance liquid chromatography (HPLC) and gas chromatography (GC) (14). These techniques require cumbersome measurement processes or rigorous measurement environments, which are destructive. In recent years, there have been some nondestructive measurement methods, such as ultrasound imaging (UI), fluorescence imaging (FI), spectroscopy and computer vision (15–17). Spectroscopy and imaging are two of the most important techniques, and the composition of the samples can be obtained by extracting the spectral data from the 380–2526 nm range. Also, imaging technique is mainly used for the acquisition of external information. The captured spectra are analyzed in conjunction with chemometric to realize the detection and discrimination of material internal information (18). The detection method has the characteristics of fast detection, high efficiency and low cost, it is suitable for long-distance acquisition and is nondestructive to samples (19, 20). Therefore, spectroscopic imaging techniques have gained great significance in the measurement of microalgae growth information detection, because they can solve some of the existing problems presented by the traditional methods and instruments.

In recent years, some studies have applied spectroscopy to the detection of microalgae life information. However, reviews for the application of spectroscopy and imaging technology for microalgae life information are lacking. Considering the important economic and medicinal value of microalgae in the world and the large amount of literature available on this area of research, a review would be valuable. During the past few years, there have been some reviews in the field of microalgae spectroscopy (21–24).

Table 1. Symbols and abbreviations used.

Abbreviation	Term	Abbreviation	Term
FTIR	Fourier transform infrared	SG	Savitzky-Golay convolution smoothing
VIS	Visible	MSC	Multiplicative scatter correction
NIR	Near infrared	SNV	Standard normal variables
HPLC	High-performance liquid chromatography	FT	Fourier transform
GC	Gas chromatography	WT	Wavelet transforms
UI	Ultrasound imaging	PLS	Partial least square
FI	Fluorescence imaging	MLR	Multiple linear regression
Chl	Chlorophyll	SVM	support vector machine
RRS	Resonance Raman	PCR	Principle component regression
SERS	Surface-enhanced Raman	ANN	Artificial neural networks
CRS	Confocal Raman	LDA	Linear discriminant analysis
CARS	Coherent anti-Stokes Raman	R ²	Coefficients of determination
SRS	Stimulated Raman	RMSE	Root mean square Error of calibration
LTRS	Optical tweezers Raman	RMSECV	Root mean square error of cross-validation
SCRS	Single-cell Raman spectra	RMSEP	Root mean square error of prediction
PAM	Pulse amplitude modulation	RPD	Residual predictive deviation
ATR	Attenuated total reflection	SPA	Successive projections algorithm
TAG	Triacylglycerol	GCV	Gross calorific value
TPEF	Two-photon excited fluorescence	PCA	Principal component analysis
UVE	Uninformative variable elimination	MCR	Multivariate curve resolution
RCF	Rolling circle filter	IMC-PLS	Interval Monte Carlo partial least squares

This article is a supplement and continuation of previous work and aims to provide a comprehensive and in-depth review of microalgae spectral detection technology. Therefore, the objective of this paper is to summarize the applications of these spectroscopic techniques for the detection of microalgae life information (with an emphasis on research after the 2010s) and discuss their advantages and shortcomings for practical applications (Table 1).

VIS and NIR spectroscopy

Visible/near infrared (VIS/NIR) spectroscopy is a noninvasive measurement technique that requires minimal or no sample pretreatment to achieve rapid detection of samples. This analysis can simultaneously achieve detection and greatly reduce the amount of required manpower, materials and time costs. Based on the characteristics of VIS and NIR spectroscopy, it is widely used in many fields, such as soil pollution detection (25, 26), food safety testing (27, 28), water pollution testing (29–31), agricultural science (32–35) and microbial detection (36, 37). The following section describes the principle of visible/near infrared spectroscopy and its application in the field of microalgae.

The principle of VIS and NIR spectroscopy

The visible spectrum has a spectral range of 380–780 nm and occupies a very small area in the entire electromagnetic spectrum. The longest wavelength is red, followed by orange, yellow, green, indigo, blue and violet. The maximum absorption in this spectral region for some particular compounds corresponds with the structure, geometry and symmetry of the material (38).

Near infrared spectrum (NIRS) is a kind of electromagnetic wave between visible spectrum and middle-infrared spectrum. Its spectral range is 780 to 2500 nm (8), which

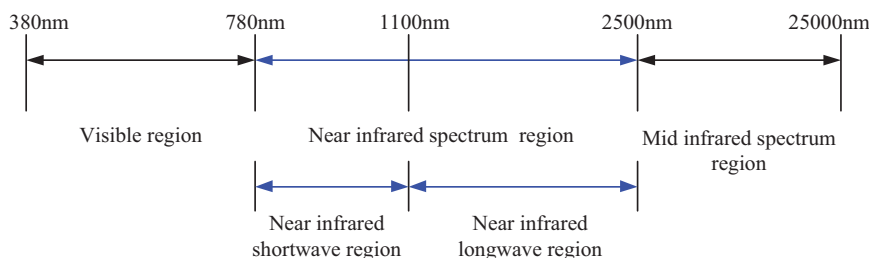


Figure 1. Spectral wavelength distribution diagram.

is the electromagnetic spectrum between visible light (VIS) and mid-infrared light (MIR) with a wavenumber of approximately $10,000$ to 4000 cm^{-1} (39), as shown in Figure 1. The overtone band and combinatorial bands of hydrogen-containing groups (C-H, O-H, N-H, S-H) in organic molecules happen to fall in the near-infrared spectral region with characteristics of absorption frequency and easy to obtain stable spectra (40). By scanning the near-infrared spectrum of the sample, the characteristic information of the hydrogen-containing groups of the organic molecules in the sample can be obtained. Therefore, NIR contains a lot of information about the structure and composition of organic compounds. Combined with modern chemometrics, it can achieve qualitative identification and quantitative analysis of complex substances (41).

Qualitative analysis by VIS and NIR spectroscopy is an important method for the calibration of material composition. Due to different chemical composition of different substances, hydrogen-containing group multiples and harmonics, the vibration frequency is different. The peak position of the spectrum, peak number and peak intensity are also different. The greater the difference in the chemical composition of the sample, the stronger the difference in the characteristics of the spectrum. Using peak identification can intuitively distinguish different substances.

Quantitative analysis using visible light near infrared spectroscopy is a widely used method. Combined with chemometrics, it is possible to quantitatively analyze the content of a substance, usually without isolation, without any chemical treatment or even with physical treatment. The content of a certain type or component in a complex sample can be obtained by spectral analysis.

The application of VIS and NIR spectroscopy

In recent years, in the field of microalgae life information detection, some researchers have started researching visible near-infrared spectroscopy. Researchers have utilized VIS and NIR spectroscopy to predict various kinds of life information in the process of microalgae culture. Table 2 summarizes the applications of VIS and NIR spectroscopy in the microalgae field in recent years. Some of the studies mentioned in Table 2 are described in detail in the following sections.

Detection of pigment

The detection of photosynthetic pigments in microalgae can be used to monitor the growth and development of microalgae. This section introduces the detection of

Table 2. Applications of VIS and NIR spectroscopy for the determination of microalgae life information.

Species	Analyte	Spectral range (nm)	Method	Evaluation parameters	Advantages	Limitations	Reference
<i>Pluvialis floto</i>	Chlorophyll a	346–1038 nm	SG-PLS	RPD = 2.6925	Two spectral preprocessing Algorithms of SG and SNV are synthetically analyzed	The reference light is not calibrated, which will increase the measurement error.	(42)
	Chlorophyll b	346–1038 nm	RAW-PLS	RPD = 1.9477			(42)
<i>Spirulina</i> sp.	Carotenoids	346–1038 nm	SPA-PLS	RPD = 3.44	In situ measurement of carotenoids	There was a lot of noise at both terminals of the spectra curve	(43)
<i>Nannochloropsis</i> sp.	Triglyceride	400–2500 nm	MSC-no Vis-PLS	RMSECV = 0.182	They verify the accuracy of prediction the combined multiple-species models	Did not consider the impact of water	(44)
	Phospholipid	400–2500 nm	MSC-no Vis-PLS	$R^2=0.969$ RMSECV = 0.227			(44)
<i>Chlorella</i> sp.	Lipids	325–1075 nm	SPA-MLR	$R^2=0.951$ RPD = 1.7537	Spectral measurements under different light condition	Did not consider the impact of water	(40)
	Fatty acids	1030–1500 nm	PLS	RPD = 4.98			(45)
<i>Chlorella</i> sp.	Palmitic acid	1600–1880 nm	PLS	RPD = 3.58	Quantification of individual fatty acids in microalgae	Other fatty acids in trace amounts were not considered	(45)
	Stearic acid	1030–1500 nm	PLS	RPD = 3.29			(45)
	Oleic acid	1030–1500 nm	PLS	RPD = 4.81			(45)
	Linoleic acid	1600–1880 nm	PLS	RPD = 2.92			(45)
	Linolenic acid	1030–1500 nm	PLS	RPD = 4.80			(45)
	Ash content	1100–2500 nm	PLSR	$r^2=0.994$			(39)
	N	1100–2500 nm	PLSR	$r^2=0.787$	Require only one spectrum, available in minutes at most, real-time measurements	Can not accurately determine lipid or total fatty acid content in ATS samples	(39)
<i>Nannochloropsis and Nannochloris</i>	Neutral lipids	1000–2500 nm	PLSR	$r^2=0.77$; RPD = 2.1	Eliminate the influence of water	High ash content, Neutral lipid value significantly lower compared to literature values	(46)
	Crude protein	1000–2500 nm	PLSR	$r^2=0.77$; RPD = 2.1			(46)
	Gross calorific	1000–2500 nm	PLSR	$r^2=0.77$; RPD = 2.12			(46)

<i>Chlorella vulgaris</i>	Ash content	1000–2500 nm	PLSR	$R^2=0.53$; RPD =1.48	A low-effort sample preparation method (dry filtrates)	The robustness of the PLSR models across growth conditions	(46)
	Biomass	300–1100 nm	MSC-PLSR	$R_{CV}=0.955$ SDR =3.38			(47)
	Lipid	300–1100 nm	MSC-PLSR	$R_{CV}=0.874$			(47)
	Lipid content	1100–2500 nm	MSC-SNV-PLS	SDR =2.06			(48)
<i>Chlorella</i> sp.	Lipid content	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.91$	The use of NIR for the full biochemical composition of microalgal biomass, using a combined species prediction model	Lack of cross-species prediction across more widely different algal species	(48)
	Carbohydrate	400 – 2500 nm	MSC-SNV-PLS	RMSEC =3.84			(48)
	Protein content	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.83$			(48)
	Lipid content	400 – 2500 nm	MSC-SNV-PLS	RMSEC =5.31			(48)
<i>Scenedesmus</i> sp.	Lipid content	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.96$	The use of NIR for the full biochemical composition of microalgal biomass, using a combined species prediction model	Lack of cross-species prediction across more widely different algal species	(48)
	Carbohydrate	400 – 2500 nm	MSC-SNV-PLS	RMSEC =2.11			(48)
	Protein content	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.91$			(48)
	Lipid content	400 – 2500 nm	MSC-SNV-PLS	RMSEC =3.84			(48)
<i>Nannochloropsis</i> sp.	Carbohydrate	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.91$	Rapid screening	For <i>Nannochloropsis</i> sp., the model metrics were less favorable	(49)
	Protein content	400 – 2500 nm	MSC-SNV-PLS	RMSEC =3.84			(49)
	Lipid content	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.91$			(49)
	Carbohydrate	400 – 2500 nm	MSC-SNV-PLS	RMSEC =3.84			(49)
<i>Kirchneriella</i> sp.	Protein content	400 – 2500 nm	MSC-SNV-PLS	$R^2=0.91$	Rapid screening	For <i>Nannochloropsis</i> sp., the model metrics were less favorable	(49)
	Biomass	1110–1900 nm	MSC-PLS	RMSEC =3.84			(49)
	Fatty acid methyl esters (FAME)	1110–1900 nm	MSC-PLS	$R^2=0.96$			(49)
	Biomass	1110–1900 nm	MSC-PLS	RPD =4.8			(49)
<i>Nannochloropsis</i> sp.	Fatty acid methyl esters (FAME)	1110–1900 nm	MSC-PLS	$R^2=0.97$	Rapid screening	For <i>Nannochloropsis</i> sp., the model metrics were less favorable	(49)
	Biomass	1110–1900 nm	MSC-PLS	RPD =5.6			(49)
	Fatty acid methyl esters (FAME)	1110–1900 nm	MSC-PLS	$R^2=0.85$			(49)
	Biomass	1110–1900 nm	MSC-PLS	RPD =2.6			(49)

photosynthetic pigments in microalgae by Visible/Near-Infrared Spectroscopy, such as chlorophyll a, chlorophyll b and carotenoids. Satisfactory experimental results confirm the feasibility of fast nondestructive detection of photosynthetic pigments.

Shao et al. (43) used VIS and NIR spectroscopy to detect the content of carotenoids in *Spirulina*. In their study, they use the spectrometer for suspension measurements in the wavelength range of 346–1038 nm. They also compared the prediction performance of x-LW-PLS, UVE-PLS and SPA-PLS. The experimental results show that the residual predictive deviation (RPD) of x-LW-PLS, UVE-PLS and SPA-PLS models were 3.40, 1.59 and 3.44, respectively. SPA-PLS gave the best result. They proved that VIS/NIR transmission spectra feasibly assessed CC of *Spirulina* sp. Combined with the successive projections algorithm (SPA), the prediction accuracy can be improved. Similarly, Zhou et al. (42) discussed the feasibility of using immersion on visible/near infrared spectroscopy for rapid detection of chlorophyll content of raw *pluvialis*. The experimental results show that the predicted effect of SPA-PLS model on chlorophyll a and chlorophyll b content was better than the full-band partial least squares (PLS) model. The RPD values of chlorophyll a and b were 2.9461 and 1.9023, respectively, indicating that the pretreatment spectrum combined with the modeling algorithm can predict the contents of chlorophyll a and b. The predictive effect of chlorophyll a was better than chlorophyll b.

The band used in both articles has a range of 346–1038 nm. This is because the absorption peaks of the carotenoids and chlorophylls, both of which are photosynthetic pigments, are within this range. The absorption spectrum of carotenoids has two adjacent absorption peaks at a wavelength of 450 nm, and chlorophyll has two absorption peaks at 450 nm and 650 nm, respectively. Different concentrations of photosynthetic pigments have different absorption capacities for the spectra, so the transmission spectra were used to quantify the photosynthetic pigments of the microalgae. Absorption spectrum that appears in the article is the spectrum of the reaction light absorption ability. When using the transmission mode for spectrum acquisition, the presence of water will have a significant effect on the spectrum of photosynthetic pigments.

Detection of lipids

The accumulation of lipids in microalgae is a guarantee for the development of microalgae energy. The rapid acquisition of the information of lipid content in the growth process of microalgae will help the farmers optimize the growth environment of microalgae, such as light, temperature, nutrient elements, etc, and make appropriate management strategies. Because lipids contain hydrogen-containing groups, information on lipids can also be expressed in the spectrum.

Lieve et al. (44) developed a multivariate calibration model based on NIR spectroscopy for predicting the content of spiked neutral and polar lipids in microalgae. In their study, they chose four samples, namely, *Nannochloropsis* sp., *Chlorococcum* sp., *Spirulina* sp. and an unknown Diatom. Spectra in the spectral region of 400–2500 nm were taken from these samples and used to create different types of PLS models, PLS1 and PLS2. The experimental results showed that NIR spectra from the four species of algae could be used to predict exogenous lipid content ($R^2=0.969$ and 0.951 and

RMECV = 0.182 and 0.227% for trilaurin and phosphatidylcholine spike, respectively). In this paper, they studied the effects of mathematical preconditioning on the quality of the predicted models and verified the accuracy of prediction of the single species models as well as the combined multiple-species models. Although the experimental results were satisfactory, the presence of water will affect the quality of the spectrum and subsequent multivariate calibration. Therefore, before applying it to unknown samples, a more powerful calibration model is needed.

Wei et al. (40) chose *Chlorella* as a research object. They used visible/near infrared spectroscopy to study the dynamic changes of fat and the distribution of fat content in the processes of microalgae growth. The detection range was 325–1075 nm, and PLS was used to analyze the spectral modeling. Successive projections algorithm was used to extract the feature bands. The results show that the model had good prediction effect ($R^2=0.940$ and RMSEP = 0.00356). In this study, rapid and nondestructive testing of the dynamic changes of fat content of *Chlorella vulgaris* was carried out by visible/near infrared spectroscopy. The results show that *Chlorella* accumulates a large amount of fat near the stationary phase of growth. The results of this study provide a method for rapid and nondestructive testing of life information, such as microalgae growth and oil content, and provide a theoretical basis and technical support for production training and strategy formulation.

The use of near infrared spectroscopy has been used to detect microalgae fatty acid content. Bin Liu et al. (45) used PLS models to analyze the individual fatty acids in microalgae. To minimize the interference caused by proteins (C-N and C=O bonds), polysaccharides (C-O bonds) and water (O-H bonds). In their experiments, the spectral acquisition ranges were 1030–1500 nm and 1600–1880 nm, and an additional CV-NIRS model was developed based on the acquired spectral data. The model had excellent performance for the prediction of TFA content and individual fatty acids (C16:0, C18:0, C18:1 and C18:3). The R^2 values were 0.998, 0.997, 0.989, 0.991 and 0.997, and the RPD values were 14.68, 10.59, 6.22, 7.31, 2.58 and 9.50. In their research, they first developed a new near-infrared spectroscopy (NIRS) technique to quickly and reliably analyze microalgae fatty acids, which can be done in seconds and requires only a small amount of sample. Compared with the traditional GC-mediated analysis, this method has great potential for high-throughput applications.

In the study of Vineela Challagulla et al. (47), four microalgae were used as experimental samples. Microalgae culture samples were filtered through GF/C filter paper, and the spectral data of wet and oven drying (60 °C overnight) filter paper were recorded in the range of 300–1100 nm and 1100–2500 nm, respectively. The experimental results showed that the spectral preprocessing did not improve the statistical data of the model. In the case of total lipid, the Partial Least Squares Regression (PLSR) calibration model developed by the NIR spectrum of dry filtrate had the same function as the prediction system of single species (*C. vulgaris*). Therefore, they recommend using the NIR region spectrum of the filtrate to estimate the total lipid content. However, under N-, P- and Fe-limited growth trials, the dry filtrate models for biomass and lipid content performed poorly in the prediction of samples. The later work in the literature suggests that further efforts should study the robustness of the restricted model.

In terms of lipids detection, the above articles are mainly on the quantitative detection of algal lipids. Different types of lipids correspond to different spectral

measurement ranges. The nondestructive testing of the sample is still guaranteed during the measurement process. Because microalgae are prone to lipid accumulation during growth, it is necessary to shake the samples before each measurement.

Detection of other analytes

The spectral analysis of other analytes includes protein content, polysaccharide content, ash content and starch content. These studies have shown only a small amount in the last five years.

Ge et al. (46) investigated the usefulness of near infrared for determining neutral lipids, crude protein, gross calorific value (GCV) and ash content in lyophilized green algae samples (*Nannochloropsis* and *Nannochloris*). To find the relationship between the chemical data in the laboratory and the absorption spectrum in NIR areas, PLS was used to calibrate and verify the model. For NIR, the prediction of crude protein was the best ($R^2 = 0.90$), followed by neutral lipids ($R^2 = 0.73$) and ash content ($R^2 = 0.70$). GCV had the lowest predictive accuracy; $R^2 = 0.57$. Thus, NIR has the potential to characterize algal biomass composition. However, the application of this technology to algae culture (99.9% water) is challenging due to the existence of water in the culture medium. The article quotes other scholars' views that Raman spectroscopy can complement NIR in order to solve the challenges posed by the water content of algae for the fast and nondestructive analysis of components (50, 51). Raman spectrum will be described in detail in the following chapters.

However, not all studies have achieved satisfactory results. Walter Mulbry et al. (39) evaluated whether NIRS could be used to determine the composition of algal turf scrubber samples. The results showed that NIRS can accurately determine ash ($R^2 = 0.994$) and total N ($R^2 = 0.787$) content but not phosphorus, total sugar, or mono-sugar content in samples. By measuring a subset of the samples, they found that NIRS cannot accurately determine lipid or total fatty acid content in ATS samples.

Based on the above results, it is clear that PLSR is one of the most commonly used and most available algorithms for multivariable analyses and model establishment. However, for different experimental parameters, the number and location of the best wavelengths in the study are different, although all of them were selected by selected by R_C of PLSR. This difference was due to the spectral absorption of different substances. Although there were cases where the experimental results were not satisfactory, most of the R_p^2 exceeded 0.8, which shows that the combination of VIS and NIR spectroscopy and chemometrics is a promising technique for detecting life information of microalgae. Selecting suitable modeling methods and spectrum preprocessing methods plays an important role in improving prediction performance.

Raman spectroscopy

Raman spectroscopy is another spectral measurement tool and it is a new field of spectroscopy. The Raman spectrum contains very rich information (20). In molecular structure analysis, Raman and infrared spectra complement each other, and some information that cannot be detected by infrared spectroscopy can be well represented in

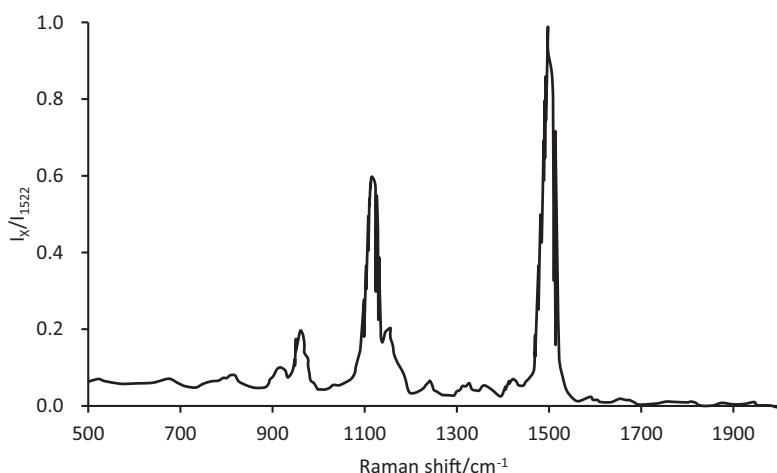


Figure 2. Raman spectrum after pretreatment (RCF, S-G Smoot-Hing, Baseline, Maximum normalization) of *Chlorella pyrenoidosa*. Reproduced from Fang et al. (56) with permission from *Spectrosc. Spect. Anal.*

Raman spectroscopy. Infrared spectroscopy focuses on the detection group, which is suitable for polar bonds, and is used to measure organic matter. Raman spectroscopy is used to detect molecular skeletons. It is suitable for nonpolar bonds and can be used to detect organic and inorganic substances simultaneously. Therefore, Raman spectroscopy can be used to monitor various molecules in microalgae. At present, most studies are concerned with the quantification of lipids and pigments, while the number of other studies is relatively small. Because of the strong Rayleigh scattering, the spontaneous Raman signal is usually weak, so it is easy to be interfered by fluorescence signal (52). Researchers have developed some techniques to enhance the signal of traditional Raman spectroscopy. Namely, Resonance Raman (RRS), surface-enhanced Raman (SERS) (53), confocal Raman (CRS), coherent anti-Stokes Raman (CARS) (54), stimulated Raman (SRS) and optical tweezers Raman (LTRS). This section describes the application of Raman spectroscopy in the detection of microalgal life information.

The principle of Raman spectroscopy

Raman spectroscopy is a kind of spectroscopic analysis technology. It is based on the Raman scattering effect found by the Indian scientist CV Raman. The scattering spectra with different frequencies of incident light are analyzed to obtain molecular vibration and rotation information. Like IR spectroscopy, in the Raman spectrum, the abscissa indicates the Raman shift and the ordinate indicates the Raman intensity, which can be used to analyze the intermolecular bond energy information (i.e., the frequency shift relative to the excitation source, expressed as wavenumber cm^{-1}) (19, 55). Some sharp peaks are often found in Raman spectra, which are characteristic of certain molecules in a sample, which makes the Raman spectrum have the function of qualitative analysis and distinguishing of similar materials. As shown in Figure 2, it is the Raman spectra of *Chlorella pyrenoidosa*. In addition, the relationship between the peak intensity of the Raman spectrum and the concentration of sample molecules is positively correlated, so

it can also be used for quantitative analysis. Due to the maturity of Raman spectroscopy and instrument manufacturing, there have been some studies using Raman techniques in various fields (57–61).

The application of Raman spectroscopy

Raman spectroscopy combines Raman and digital imaging and provides spectral and spatial information, which expands the scope of application of Raman technology (62). When Raman spectroscopy is used to detect microbiological cells, it can obtain a complete Raman spectrum at different locations on a sample and present a three-dimensional image with high spatial resolution and weak background interference (63). Consequently, it can be used well in biological samples (64, 65).

Regarding the study of Raman spectroscopy in microalgae, Samek et al. (50) reviewed the current status of Raman spectroscopy applications on algal research in 2011 (in particular its use for the determination of important parameters of fatty acids). Due to its early stage of development, the number of publications covering the topic and the number of research groups devoted to this field are still deficient. Since 2011, the application of Raman spectroscopy in microalgae has been extensively reported. Therefore, this section reviewed the application of Raman spectroscopy for the detection of microalgae life information after 2011. We discussed the advantages and disadvantages of each work in this section and summarized the results in Table 3.

Application of Raman spectroscopy for lipids

Since 2010, the interest of Raman spectroscopy in algal research has increased. Most studies focused on the characterization of lipid/fatty acids in microalgae by Raman spectroscopy. The large proportion of "C=C" and "C=O" in the lipid structure has a great contribution to the molecular vibration in the Raman spectrum, and its information can be clearly expressed in the Raman spectrum.

Fang et al. (56) also studied the changes of microalgae lipid under Nitrogen-Based Stress by using Raman microscopy. They used Raman spectroscopy to obtain Raman spectra of *Chlorella* grown under three conditions: nitrogen stress, nitrogen saturation and nitrogen saturation. The growth of microalgae and changes of lipid under different nitrogen stresses were studied by spectroscopic analysis. After spectrum signal preprocessing, principal component analysis (PCA) was used to analyze the whole band to obtain the corresponding principal component variables, and the classification model was established by linear discriminant analysis (LDA). The forecasting accuracy of the three nitrogen nutrient conditions was 80, 93.3 and 86.7%, respectively. The classification model established by the ratio of oil feature displacement (RS) maximized the classification accuracy of the three nitrogen nutrient conditions to 86.7%. They also confirmed that the use of Raman technology to differentiate the different nitrogen stress conditions of microalgae growth is feasible, and with the increase of the time of nitrogen stress, there was a greater difference in oil accumulation.

Phiranuphon Meksiarun et al. (73) introduced a noninvasive, quantitative analysis technique. Using living marine diatoms as the research object, they used Raman

Table 3. Applications of Raman spectroscopy for determination of microalgae life information.

Species	Analyte	Spectral range (nm)	Method	Evaluation parameters	Advantages	Limitations	Reference
<i>Nannochloropsis oceanica</i>	Triacylglycerol	3256 cm ⁻¹ –273 cm ⁻¹	PCA-LDA-PLSR	$R^2=0.9790$	Real-time monitoring; no need to label cells	Misclassification rates at the earlier time points were relatively higher	(66)
<i>Chlorella pyrenoidosa</i>	Lipid	100 cm ⁻¹ –2000 cm ⁻¹	RCF-PCA-LDA	86.7%	Identification of different nitrogen stress conditions	Complexity of sample preparation	(56)
<i>Monoraphidium neglectum</i>	Lipid droplets	–	–	–	Separate the CARS signal from chlorophyll fluorescence	Complexity of sample preparation	(54)
<i>Chlorella</i> sp.	Species identification	500 cm ⁻¹ –2000 cm ⁻¹	PLS	80%	High prediction accuracy	Complexity of data operation	(67)
<i>Chlamydo-monas</i> sp.	Species identification	500 cm ⁻¹ –2000 cm ⁻¹	PLS	90.3%	–	–	(67)
<i>Nostoc Commune</i>	Carotenoids	900 cm ⁻¹ –1700 cm ⁻¹	–	–	The TPEF was completely blocked	Excessive exposure time can cause photodamage of carotenoids	(68)
<i>Nostoc</i> sp.	–	–	–	–	–	–	–
<i>Chlorella</i> sp.	Lipids (TAG)	2840 cm ⁻¹ –2950 cm ⁻¹	CARS imaging	–	Real-time imaging	Overlap between the fluorescence of chlorophyll and the CARS signal of lipid	(69)
<i>Coccomyxa</i> sp.	Carotenoid	1505 cm ⁻¹ –1535 cm ⁻¹	–	–	Fast speed	–	(69)
<i>S. obliquus</i>	Lipid	1445 cm ⁻¹	PCA-LDA	$R^2=0.83$	High resolution	Presence of fluorescence signals	(70)
<i>Chlorella protothecoides</i>	Glucose	500 cm ⁻¹ –4500 cm ⁻¹	RCF-PLS-SG	$R^2=0.9734$	Real-time estimation	Complexity of data operation	(52)
<i>Chlamydomonas reinhardtii</i>	Starch	394.1 cm ⁻¹ –3340.9 cm ⁻¹	SCRS	$R^2=0.9893$	Prediction performance improvement	–	–
<i>Dunaliella</i>	Algal bioactive	251 cm ⁻¹ –1999 cm ⁻¹	PCA	–	Rapid, label-free, noninvasive, culture-independent, low-cost	Wide adoption can be limited, Other glycosyl polymers interfere	(71)
<i>Chlamydomonas reinhardtii</i>	Starch content	393.8 cm ⁻¹ –1801.4 cm ⁻¹	PLSR	$R^2=0.9855$	Without additional fluorescence labeling, high sensitivity	The model is sensitive to spectral background	(53)
<i>Chlamydomonas reinhardtii</i>	Protein content	2701.6 cm ⁻¹ –3051.6 cm ⁻¹	PLSR	$R^2=0.9864$	Widely applicable nondestructive	acquisition cannot be performed onsite	(72)
<i>Chlamydomonas reinhardtii</i>	Triacylglycerol content	2701.6 cm ⁻¹ –3051.6 cm ⁻¹	PLSR	$R^2=0.9318$	–	–	(72)
<i>Thalassiosira (T.) pseudonana</i>	Lipid	393.8 cm ⁻¹ –1801.4 cm ⁻¹	PLSR	–	Noninvasive non-preprocessing	Sufficient data pretreatment	(73)
<i>E. gracilis</i>	D ₂ O	730 cm ⁻¹ –1800 cm ⁻¹	PLS	–	No background Raman signal	Need air-dry	(74)
<i>Botryococcus braunii</i>	Hydrocarbons	700 cm ⁻¹ –3200 cm ⁻¹	–	–	Innovative application	Spectrum is not refined enough	(75)
<i>Aurantiochytrium mangrovei</i>	Squalene	200 cm ⁻¹ –3600 cm ⁻¹	–	–	Label-free molecular imaging, live detection	Complex signal data processing	(76)
<i>Chlorella vulgaris</i>	Lipid Carotenoids	100 cm ⁻¹ –3000 cm ⁻¹	–	–	–	–	(76)
		2845 cm ⁻¹ –3107 cm ⁻¹	–	–	–	–	(77)

Spectroscopy combined with the Orthogonal Least Squares (OLS) method to assess fat composition in samples without special treatment. Spectral data of accumulated fatty acids were obtained in the spectral region of 800–1200 cm^{-1} , and the real-time changes of individual lipids in diatom cells were observed. The experimental results show that the correlation coefficient of OLS analysis was above 0.9, and the average Raman data was consistent with the gas chromatographic analysis. This also shows that the analysis of total lipids in diatoms using Raman spectroscopy is reliable.

Many researchers have used Raman microspectroscopy, to capture Raman spectra of samples at a single cell level. This is a hot spot of research in recent years. Tingting Wang et al. (66) used Raman microspectroscopy to establish the quantitative dynamics of triacylglycerol accumulation in microalgae population. In this study, single *Nannochloropsis oceanica* cells were captured by a 532-nm laser, and the Single-cell Raman spectra (SCRS) were acquired by the same laser within one second per cell. The accuracy of the differentiation of nitrogen-replete cells or nitrogen-depleted cells within 6 h was more than 93.3% ($>93.3\%$), but the accuracy was $>90.4\%$ among the eight time points under nitrogen depletion. They demonstrated that SCRS can be used to quantitatively assess the TAG content and lipid saturation of individual microalgae. It also enabled rapid, quantitative and noninvasive tracking of triglyceride dynamics at the single-cell level in real time.

Daniel Jaeger et al. (54) used *Monoraphidium neglectum* as a research object, applied CARS microscopy to analyze algal lipids, and performed *in vivo* imaging of lipid content in a label-free and nondestructive manner. A method to separate the CARS signal from the two-photon excited chlorophyll fluorescence using a modulation scheme was proposed. Lipid droplet formation during nitrogen starvation was directly analyzed for the first time. This work demonstrated the potential of quantitative CARS microscopy as a marker free analysis technology for microalgae lipids, which elucidates the process of lipid accumulation in microalgae and is highly correlated with future biotechnological applications.

Shao et al. (70) chose *Scenedesmus obliquus* as an experimental object. Under the condition of nitrogen stress, the lipid content in microalgae was qualitatively and quantitatively analyzed by confocal Raman microspectroscopy. The spectral signals from different microalgal cells were collected in order to establish a prediction model of lipid content in cells, rendering surface scanning signal and analysis of Raman spectrum data. The PCA-LDA model was established to classify the collected data from different periods of microalgae, and the correct classification rate was 100%. The experimental results also showed a significant correlation ($R^2 = 0.83$) between the results of the correlation analysis of the total lipid content determined by the gravimetric method and the correlation analysis of the Raman intensity (1445 cm^{-1}) of the characteristic peak of lipids. The spectroscopic method can be applied to semi-quantitatively detect the lipid content of microalgae.

He et al. (69) used *Coccomyxa* sp. C169 as the research object and used spontaneous Raman scattering and coherent anti-Stokes Raman scattering (CARS) spectroscopy and microscopy to monitor the growth of triacylglycerols (TAGs) in microalgal cells under the condition of N-depletion. *Coccomyxa* accumulated a large amount of TAG under the conditions of nitrogen depletion (N-depletion). In this study, CARS spectra were

compared with Raman spectra. The results showed that the Raman spectra had much better spectral resolution than B-CARS spectra. While CARS imaging of microalgae had the disadvantages of signal overlap with two-photon excited fluorescence, it was more robust than Raman imaging in terms of speed and spatial resolution. Subsequent studies will focus on reducing the non-resonant background of CARS microscopy and real-time monitoring of microalgal growth in N-depleted environments.

Mats Josefson et al. (53) listed an example of how to use T-OPLS to evaluate resonant Raman spectra. In the example, the spectral range was set to $251\text{--}1999\text{ cm}^{-1}$. Using PCA, a model was established to determine the distribution of natural products in complex matrices, such as algae. The results show that surface-enhanced Raman spectroscopy (SERS) can image individual algae cells to identify and determine the spatial distribution of individual compounds.

Kei Ishitsuka et al. (76) performed label-free molecular imaging using a hyperspectral CARS microscope, successfully visualized the molecular species in living algae, and successfully performed the classification and identification of intracellular lipids of *Aurantiochytrium mangrovei* strain 18W-13a.

The above results show that Raman spectroscopy can be used as a new detection technology to provide sufficient phenotypic information for the quantitative evaluation of its lipid content in microalgae. But the researchers realized that although the presence of water has a very small impact on the Raman spectrum, the strong fluorescence makes the obtained Raman spectrum complicated, which may mask the characteristics of the Raman spectrum. Signal preprocessing technology, such as rolling circle filter (RCF), can be used as a method to remove fluorescence background in Raman spectra.

Application of Raman for the identification of algae species

Raman spectroscopy is a spectroscopy tool that is used to monitor a variety of molecules in microalgae in which most studies are aimed at the quantitative determination of various lipids and pigments (23). Raman spectroscopy can also be applied to the detection of other compounds in microalgae and identify microalgae species. However, research related to these aspects is scarce. A preliminary feasibility study of using Raman spectroscopy to classify algae species was carried out in 2015. Shao et al. (67) believes that the identification of the biomolecules in microalgae cells by Raman spectroscopy is the basis for the identification of microalgae species. Because the characteristic peaks of the Raman spectrum of β -carotene are different in different microalgae, they used Raman microspectroscopy technology in combination with PLS full band modeling to study the possibility of rapid identification of *Chlorella* sp. and *Chlamydomonas* sp. To solve the problem of the variation of Raman spectra in different conditions for the same algae species, the maximum spectral-to-peak ratio method was proposed to eliminate the intensity variation of single spectral peaks. The article compares the differentiating effects of PLS modeling under different pretreatment methods. The results showed that when the threshold value was ± 0.2 , the accuracy of prediction for the two algae was 80 and 93.3%, respectively. When the threshold value was ± 0.5 , the accuracy of prediction for both algae reached 100%. The results show that the two

microalgae classifications can be distinguished by this method, which provides a new idea for algae species identification and algal species classification research.

Raman in other aspects of the application of microalgae

The other life information of microalgae includes protein content, glucose content, carotenoid content and starch content. The number of publications based on the Raman spectrum of these life information and the number of research teams involved in the field are still very few so far, only a small amount of these studies appeared after 2010.

He et al. (72) established a method based on SCRS and simultaneously quantified the content of starch, protein, three acylglycerols and lipid unsaturation in individual cells. By deep sampling the SCRS of *Chlamydomonas reinhardtii* at 16 time points over 8 days under nitrogen depletion, the authors developed a full spectral prediction model based on PLSR. The prediction accuracy of all models is 96.86–99.24%, which is significantly higher than that of unimodal models. Because the accuracy and reliability of the measurement depends on the number of cells sampled, they proposed a formal mathematical framework and rationally defined “minimal sampling depth”. With the appropriate sampling depth, SCRS can be used as a quantitative and qualitative tool for the characterization and screening of macromolecules in algae.

Andrej Dementjev et al. (68) used coherent anti-Stokes Raman scattering microscopy for imaging experiments. Three types of algae were used as research objects. In the imaging experiments, the spectral characteristics of carotenoids were selected as markers. The experimental results show that the generation of strong two-photon excited fluorescence (TPEF) signals in microalgae affected the experimental results. In this paper, the use of wavelength mixing (Optical Parametric Oscillation (OPO) signal beam and basic beam) is proposed to effectively suppress TPEF signal. A high-resolution image of the distribution of carotenoids in algae and cyanobacteria was obtained experimentally, confirming the CARS microscope's ability to spatially characterize carotenoids stored in microalgal cells.

Ji et al. (71) introduced SCRS to quantify the amount of starch in living cells. In this work, *C. reinhardtii* was used as the research object, and SCRS was used to capture Raman spectra of individual single cells under different culture conditions and at different growth stages. The Raman spectrum collection range was $394.1\text{--}3340.9\text{ cm}^{-1}$. The results showed a linear correlation between the signal intensity at 478 cm^{-1} and the starch content of the cells ($R^2=0.9893$). This is a quantitative measurement of starch content at the single living cell level. This method can prevent cells from being damaged, and cells can survive after SCRS measurements.

Other aspects of Raman spectroscopy in microalgae research include using Raman spectroscopy for real-time estimation of glucose concentration under mixotrophic conditions (52). Yusuke Yonamine et al. (74) presented a method for monitoring the photosynthetic activity of *Euglena gracilis* as the model organism at single-cell resolution by Raman spectroscopy with deuterium from deuterium oxide (D_2O) as a tracking probe.

Weiss et al. (75) used Raman spectroscopy to characterize the structure of botryococcenes produced by *Botryococcus braunii* and identified and localized botryococcenes in *B. braunii* cells. The spectral range they chose was $1600\text{--}1700\text{ cm}^{-1}$. In the study, they identified specific Raman spectral features of *B. braunii* B *Staphylococcus*, and used

confocal Raman microspectroscopy to map the presence and location of methylated botryococcenes within a colony of *B. braunii* cells. This study demonstrates that Raman spectroscopy can be applied to analyze the level and quality of staphylococci during the development of *B. braunii* cultures to determine when the oil content has both the maximum number and quality for cell harvest.

In recent years, with the continuous improvement and promotion of Raman technology, algae researchers have begun to discover the superiority of Raman technology and pay more attention to this technology. The above-mentioned studies indicate that Raman spectroscopy has been widely used as a novel spectroscopic detection technique in the analysis of physical and chemical structures of materials. This technology can achieve fast and nondestructive detection of microalgal cells. It is one of the technologies available for analyzing microalgal cell components. Because the Raman spectrum is extremely insensitive to polar substances, such as water, it has good application prospects in the research of microalgal life information detection. However, there are many deficiencies in the application of Raman technology to the detection of life information of microalgae, such as the background elimination of Raman spectra and the processing of Raman data. Therefore, more research is needed to explore Raman technology.

Fourier transform infrared (FTIR) spectroscopy

Fourier transform infrared spectroscopy (FTIR) is an analytical technology for measuring infrared spectra by measuring interferogram and Fourier transform to the interferogram. There is a Fourier transform functional relationship between the intensity $h(\delta)$ of the infrared spectrum and the optical path difference between the two beams of light that form the light. FT-IR spectroscopy in the mid-infrared region (4000 to 400 cm^{-1}) is a biochemical fingerprinting technique (20). Fourier transform infrared spectroscopy is a fast, simple, sensitive and reproducible analytical technology. It is widely used in various fields to collect qualitative and quantitative analysis of material properties by collecting the internal information contained in the infrared spectrum, such as food testing (78–81), drug testing (82–84), microorganisms (85, 86) and disease detection (87, 88). This part of the literature mainly introduces the principle of FTIR spectroscopy and its application in the field of microalgae life information detection.

The principle of Fourier Transform Infrared (FTIR) spectroscopy

Fourier transform infrared is the preferred method for infrared spectroscopy. Fourier transform infrared has the following four sampling techniques: transmission; attenuated total reflection (ATR); specular reflections; and diffuse reflections. Each technology has its own characteristics, which makes them suitable for different sample conditions. When a continuous-wavelength infrared light source illuminates the sample, the molecules in the sample absorb certain wavelengths of light, and the unabsorbed light reaches the detector. The analog signal acquired by the detector is subjected to analog-digital conversion and Fourier transform to obtain a single-beam spectrum with sample information and background information. Then, the same detection method is used to obtain the background single beam spectrum where the infrared light does not pass

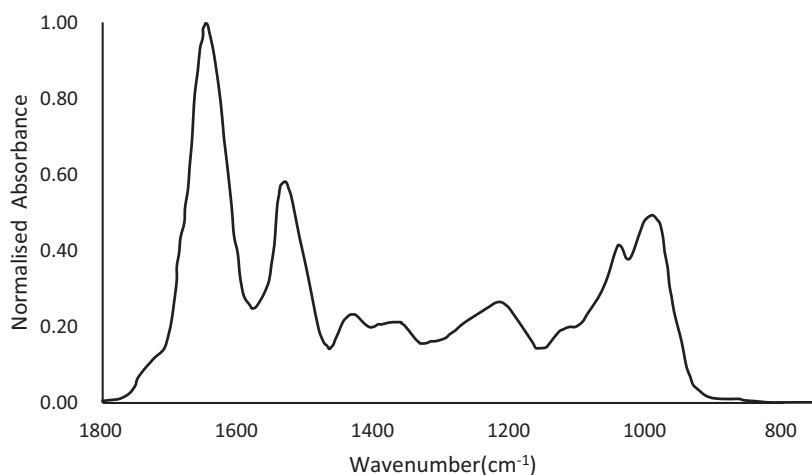


Figure 3. Averaged spectra of *Microcystis aeruginosa*. Reproduced from Kansiz et al. (91) with permission from Elsevier Science Ltd.

through the sample. By subtracting the background single beam spectrum from the single beam spectrum of the sample, an infrared “fingerprint” spectrum representing the structural characteristics of the sample molecule is generated. Because different chemical structures (molecules) produce different fingerprint spectra, organic macromolecules consist of several chemical bonds, so the spectrum of this molecule is relatively complex (89,90) (Figure 3), which depicts the FTIR spectra of *Microcystis aeruginosa* in the range of 1800–800 cm^{-1} (91).

The application of Fourier Transform Infrared (FTIR) spectroscopy

Fourier transform infrared has been extensively applied to the analysis of multiple parameters of microalgae. Using these spectral properties, Kansiz et al. (91) showed, for the first time, that FTIR spectroscopy can be applied to phototrophs to discriminate between cyanobacterial strains. This article reviews the applications of Fourier technology in microalgae-related fields since 2010. Table 4 summarizes these studies. Some of the studies mentioned in Table 3 are described in detail in the following sections. In this part, the article divides the following studies into three categories: detection of life information in microalgae, comparison and combination with other technologies and other applications.

Application of FTIR for the detection of life information in microalgae

Because all the macromolecules show unique bands in the spectra of the collected microalgae, some scholars have used FTIR spectroscopy techniques to establish a prediction model for macromolecule content in microalgae cells by measuring the absorbance of microalgae samples. The main infrared absorption bands of microalgae are shown in Table 5.

In the literature, two spectral regions are commonly used for the detection of lipid content, 2800–3000 cm^{-1} and 1740 cm^{-1} , respectively. These two spectral regions

Table 4. Applications of Fourier transform infrared spectroscopy for determination of microalgae life information.

Species	Analyte	Spectral range (nm)	Method	Evaluation parameters	Advantages	Limitations	Reference
<i>Ditylum brightwellii</i>	–	900–4000 cm ⁻¹	PLS-LDA	–	Direct estimate	Sample preparation and drying	(92)
<i>C. simplex</i>	Carbon	3050–2800 cm ⁻¹	PLSR	RPD = 9.8	Snapshot prediction	Measurements are based on a single species of cultured alga	(93)
	Proteins	1540–1000 cm ⁻¹	PLSR	RPD = 9.2			(93)
	Productivity	3050–2800 cm ⁻¹ 1540–1000 cm ⁻¹	PLSR	RPD = 13.2			(93)
<i>Nannochloropsis</i> sp.	Lipids	3050–2800 cm ⁻¹ 1540–1000 cm ⁻¹	PLS	R ² = 0.995 RPD = 8.95	High predictive accuracy, low relative error,	Sample preparation needs to be accelerated	(94)
	Proteins	4000–450 cm ⁻¹	PLS	R ² = 0.969 RPD = 5.29	quick analysis	No guarantee of real-time monitoring	(94)
	Carbohydrates	4000–450 cm ⁻¹	PLS	R ² = 0.965 RPD = 3.44			(94)
<i>Chlamydomonas reinhardtii</i>	Lipid	1800–950 cm ⁻¹	PCA	R ² = 0.9301	Whole organism analysis, using intact cells	Complex experiment environment	(95)
	Carbohydrate	1800–950 cm ⁻¹	PCA	R ² = 0.8358			(95)
	Lipid	1800–950 cm ⁻¹	PCA	R ² = 0.9002			(95)
<i>Nannochloropsis</i> sp.	Carbohydrate	1800–950 cm ⁻¹	PCA	R ² = 0.9296	Nondestructive, fast analytical, require only very small amounts of biomass, the large number of samples that can be analyzed and screened for specific characteristics	Spectral complexity, The presence of water will affect the quality of the spectra, High error rate for cross species prediction	(95)
	Triglyceride	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.944			(44)
	Phospholipid	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.682			(44)
<i>Chlorococcum</i> sp.	Triglyceride	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.902			(44)
	Phospholipid	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.801			(44)
	Triglyceride	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.920			(44)
<i>Spirulina</i> sp.	Phospholipid	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.740	Innovative sample preparation, strong applicability	Additional analytes need to be incorporated into the database	(44)
	Triglyceride	4000–500 cm ⁻¹	PLS-MSC	R ² = 0.933			(44)
	Phospholipid Compounds	4000–500 cm ⁻¹ 3940–850 cm ⁻¹	PLS-MSC PCA	R ² = 0.699			(96)
<i>Microalgae</i>				–			
<i>oculata</i>	Lipid	4000–400 cm ⁻¹	PLSR	R ² = 0.9966	Small volume sampling, rapid, robust, accurate	Model was limited to only one kind of algae	(97)
<i>Chlamydomonas reinhardtii</i>	Protein	3100–2800 cm ⁻¹	PLS	R ² = 0.994	Simultaneous and quantitative determination, without specific cell preparations	Only applies to one type of microalgae	(98)
	Carbohydrate	3100–2800 cm ⁻¹	PLS	R ² = 0.997			(98)
	Lipid	3100–2800 cm ⁻¹	PLS	R ² = 0.977			(98)
<i>Cyanobacterial</i>	Classification Macromolecule	1800–950 cm ⁻¹	SIMCA-PCA	>93%	Fast, reproducible, automated assortment	Spectra obtained showed shifting of the spectral baseline	(99)
<i>Haematococcus pluvialis</i>	Astaxanthin	4000–900 cm ⁻¹	PCA	–	Technology fusion, convenient identify the individual pure components	Advance the screening time	(100)

Table 5. Range of major FT-IR band assignments of algal.

Wavenumber range (cm ⁻¹)	Assignments	Functional groups
3400–3300	ν O–H/ ν N–H	Water, protein
3000–2800	ν (C–H)	Lipids
1740	ν C=O of esters	Membrane lipids, fatty acids
1655–1638	ν C=O	Protein (Amide I)
1545–1540	δ N–H, ν C–N	Protein (Amide II)
1514	ν C=C	carotenoid
1465–1470	δ CH ₂	lipids
1398–1370	δ CH ₃ , δ CH ₂ / δ C–O	Proteins, carboxylic groups
1320	ν C–H, δ N–H	Proteins
1050	ν_s CO–O–C	Carbohydrates
1200–950	ν C–O–O	polysaccharides
940	C–O	Carotenoid

ν = symmetric stretching, ν_{as} = asymmetrical stretching, δ = symmetric deformation (bend), δ_{as} = asymmetric deformation (bend). Band assignments are taken from references (22, 89, 94, 95, 97, 98, 101).

represent vibration stretching of methyl and methylene groups and the vibrational stretching of the C=O bond, respectively (94). As shown in Table 5, the amide I (1655 cm⁻¹) band and amide II (1545 cm⁻¹) band can be used as two regions of quantitative protein content. Pistorius et al. (102) believe that the quantitative detection of protein by amides II is more reliable. Because of the complex composition of carbohydrates, its absorption spectrum is composed of complex overlapping bands between 1200 and 1000 cm⁻¹. However, Wagner et al. (98) found that the obvious characteristic peaks at 1050 cm⁻¹ have a good correlation with the carbohydrate content in the biomass samples.

In the study of Joshua et al. (94), they considered that the linear correlation coefficient between the vibrational stretching of ester bonds and lipid content was higher than the methyl and methylene groups. They used *Nannochloropsis* sp. as a research object and utilized FTIR spectroscopy for the simultaneous quantitative determination of lipid, protein and carbohydrate content in microalgal samples. They constructed a robust single species model. The results of the study show that the model had good predictive performance for the content of lipids, carbohydrates and proteins ($R^2 = 0.995$, 0.969 and 0.965 and RPD = 8.95, 5.29 and 3.44 for lipid, protein and carbohydrate, respectively) with relative errors below 5%.

Previous studies (101) have shown that the N-limitation has little effect on the protein content of cells, so the spectral bands of 1740 cm⁻¹ and the 1200–950 cm⁻¹ region can be used as spectrometric measurements of lipids and carbohydrates under the condition of N-limitation.

Dean et al. (95) used the FTIR to study the changes in lipid and carbohydrate content in *C. reinhardtii* and *Scenedesmus subspicatus* under N-limitation. The spectral region they choose is 1800–950 cm⁻¹. The results show that lipid and carbohydrate bands showed rapid and significant changes under intermediate-N and low-N conditions. They confirmed that FTIR can be used as an effective tool to identify lipid and carbohydrate contents in microalgae at different growth stages.

Rémy Coat et al. (97) used *Nannochloropsis oculata* as a research object and developed a novel cell sampling and monitoring method combined with detailed chromatographic analysis to monitor the heterogeneity of fresh microalgae cells. They chose the absorbance of C=O band located in 1740 cm⁻¹ as the characteristic of total lipid

content in cells. The quantification model of lipids was established using FTIR spectroscopy combined with PLS-R method. The experimental results show the FTIR model could effectively detect the total lipid content of cells ($R^2 > 98.8$ and $RMSE < 1.32$). They also developed a photobioprocess monitoring pipeline to validate their unbiased FTIR approach.

As a similar study, Heiko Wagner et al. (98) quantitatively studied the major macromolecules in *C. reinhardtii*, such as proteins, carbohydrates and lipids. They considered that the absorption of OH bonds was overlapped within the range of $3100\text{--}2800\text{ cm}^{-1}$, so the experimental spectrum ranged from 1900 to 950 cm^{-1} . The regression equation was established by PLS. The results are similar to those obtained by biochemical methods.

Olivia Sackett et al. (93) used FTIR techniques and combined chemometrics to establish a PLSR model and developed a simple linear model for snapshot predicting the carbon and protein content of diatom. They used *C. simplex* as an experimental sample. To avoid spurious absorbance values of amide I band (1650 cm^{-1}), region $3050\text{--}2800$ and $1540\text{--}1000\text{ cm}^{-1}$ were selected for spectroscopic analysis. Experimental results show that the prediction ability of carbon, protein and productivity models were all very good. The RPD showed that all models presented were of “Excellent” quality.

Horton Rebecca et al. (96) made use of innovative sample preparation and chemometric methods to overcome the difficulty of obtaining detailed information of organisms. They used FTIR spectroscopy and improved principal component regression (PCR) to study the ability of microalgae cells to respond to environmental changes. The experimental spectrum ranged from 3940 to 850 cm^{-1} , they established a new type of virtual chemometrics laboratory and connected a graphical user interface to an SQL database, enabling non-stoichiometric scientists to apply these algorithms through a graphical user interface for quantitative calibration.

Comparison and combination with other technologies

In comparison experiments between FTIR spectroscopy and other measurement technologies, Lieve (44) compared the use of NIR and FTIR to predict the amount of spiked neutral and polar lipids in microalgae. Both NIR and FTIR are nondestructive and fast analysis methods. The near-infrared absorption of chemical structure comes from the overtone and combination of vibration. They also play a significant role in FTIR spectroscopy. In NIR spectroscopy, these bands are wider and less defined. FTIR spectrum is more complex than NIR spectrum. It can provide more details and structural information of chemical structures. The results of model predictions are $R^2 = 0.969$ and $R^2 = 0.907$. The prediction performance of NIR was more accurate and robust than FTIR. The reason for this difference may be that the FTIR spectrum is more complex with more substance information in the spectrum. Thus, the robustness of the model was poor.

Both FTIR and Raman spectroscopy are vibrational spectra that are label-free, non-destructive and fast-to-use. The infrared spectrum contains the spectral contributions of macromolecules such as proteins, nucleic acids, carbohydrates, bio-silicon, and lipids. One of its drawbacks is the need to dry the sample, which can seriously affect the cell

physiology. Raman spectroscopy can be applied to water environment. The Raman spectrum of microalgae is mainly affected by photosynthetic pigments, chlorophyll a and carotenoids. Therefore, Raman spectroscopy can provide additional information for FTIR spectroscopy. In order to compare the application of FTIR and Raman spectroscopy in monitoring the growth of microalgae, Jan Rüger et al. (92) used diatom *Ditylum brightwellii* as a model and applied FTIR and Raman spectroscopy to study the chemical changes of samples during growth. The results show that the model based on Raman spectroscopy was more stable and accurate than the model based on FTIR spectroscopy. However, the article also confirmed that FTIR spectroscopy is a reliable technique that can evaluate microalgal cell physiology under controlled conditions because it provides a direct estimate of the macromolecular cell composition. Therefore, it is recommended to use the two technologies together.

Other applications

Somchanh Bounphanmy et al. (99) used filamentous *cyanobacteria* as the research object and used ATR FTIR spectroscopy combined with PCA and soft independent simulation analogy (SIMCA) to distinguish filamentous cyanobacteria from soil. The range of spectral regions used in the study was $1800\text{--}950\text{ cm}^{-1}$, and the accuracy of the classification model was better than 93%. The results showed that the ATR/FTIR spectrometry combined with the stoichiometric classification method constituted a fast, reproducible and possibly automated assortment of filamentous cyanobacteria in the soil.

Using *Haematococcus pluvialis* as an experimental sample, Liu et al. (100) screened algae strains producing astaxanthin using FTIR methods combined with chemometric methods, such as PCA and multivariate curve resolution (MCR) algorithms. Spectral collection in the experiment ranged from 4000 cm^{-1} to 900 cm^{-1} . The ratio of the 1740 cm^{-1} band to the 1156 cm^{-1} band in FTIR spectra can be used for the identification of Astaxanthin-Hyperproducing strains. For the analysis of a large number of samples, the PCA method shows a good ability to distinguish different strains of algae. This work has opened up a new way for high-throughput microbiological screening.

Based on these results, it has been suggested that FTIR spectroscopy in combination with chemometric analysis is a promising technique for the detection of microalgae life information. The proposed FTIR technology has been proven to be a fast, accurate and highly reproducible method. This method can quantify the biochemical changes during the growth of microalgae and monitor the growth and life information of microalgae. More importantly, the selection of the appropriate modeling methods and spectral pre-processing methods can effectively improve the prediction performance.

Fluorescence spectroscopy

When a substance is excited by a certain wavelength of light, it emits light with a wavelength longer than the excitation wavelength in a very short time (10–8 s). This light is called fluorescence. If the energy-wavelength diagram of the fluorescence is taken out, the diagram is the fluorescence spectrum. The fluorescence spectrum can be obtained by

spectral detection. The application of this luminescence phenomenon in various aspects is called fluorescence technology. When the object is excited, there are two types of fluorescence, namely, blue-green fluorescence and chlorophyll fluorescence derived from cinnamic acid and chlorophyll, and their spectral ranges are 400–600 nm and 650–850 nm, respectively. Among them, chlorophyll is the main fluorescent substance in plant tissues, and its photosynthetic activity is very sensitive to stress factors (62). A large number of studies have shown that the fluorescence signal emitted from the plant contains very rich photosynthesis information and is very sensitive to changes in the plant itself and the external environment and can be used for quantitative determination of substances. Therefore, fluorescence analysis technology has been widely used in various research fields as a rapid and noninvasive research method for studying plant photosynthesis processes (103, 104).

This part of the article mainly introduces the principle of fluorescence technology and its application in the field of microalgae life information detection.

The principle of fluorescence spectroscopy

Fluorescence is a kind of photoluminescence phenomenon. After a material molecule is subjected to electromagnetic radiation, its electrons will jump from the ground state to different vibrational energy levels of multiple electron singlet states, and then generate multiple absorption bands. This process takes about 10^{-15} s. The electrons in the excited singlet state are unstable. In the process of backstepping to ground state, electrons in the vibrational energy levels above the S_2 singlet state will quickly lose the excess energy, decay to the lowest excitation vibrational level of the S_1 state, and finally jump back to the vibrational energy layers of the ground state and emit fluorescence. Figure 4 shows the principle of fluorescence.

The fluorescence spectrum includes both excitation and emission spectra. Both of these spectra can be used to identify fluorescent compounds and can also be used as a direct basis for selecting the appropriate excitation and emission wavelengths for performing fluorescence measurements.

The application of fluorescence spectroscopy

Fluorescence detection is a noninvasive technology. It has the characteristics of highly sensitive, suitable for *in situ* and on-line monitoring. It has been widely applied to phytoplankton pigment composition and community distribution measurement. The use of fluorescence spectroscopy to identify phytoplankton species began in the 1980s. In 1985, Yentsch et al. (103), first used fluorescence spectroscopy to identify phytoplankton pigment species. In 1993, Cowles et al. (105) deduced phytoplankton species and abundance changes in seawater by measuring the characteristic fluorescence of phycoerythrin. In 1995, Lee et al. (106) established a method of fluorescence analysis for cyanobacterial content based on the characteristic fluorescence of phycocyanin.

Table 6 summarizes the studies that were used in the field of microalgae in the past 10 years. Some of the studies mentioned in Table 6 are described in detail in the following sections. In this part, the article divides the following studies into four categories: Fluorescence imaging, Chlorophyll fluorescence, (3D) fluorescence and other applications.

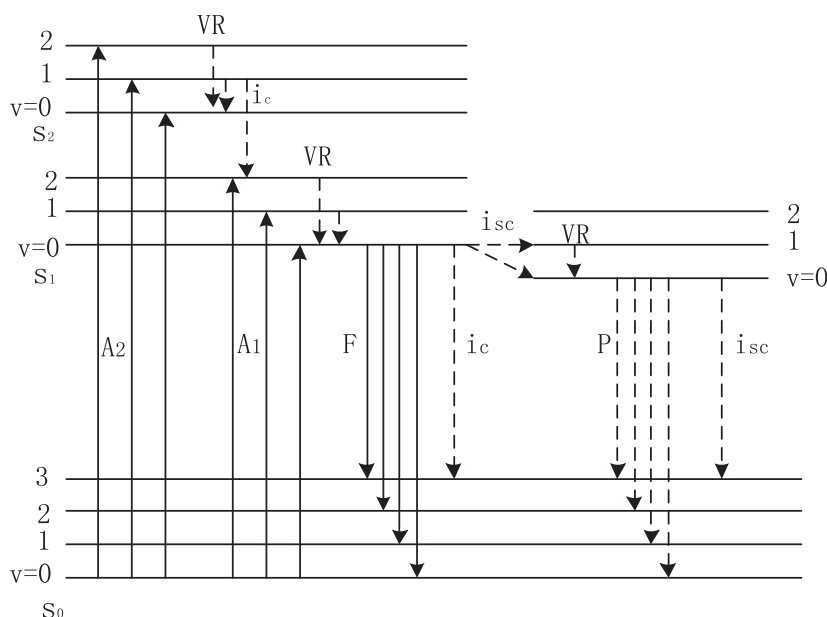


Figure 4. Intramolecular excitation and decay process. A1, A2: Absorption; F: Fluorescence; P: Phosphorescence; ic: Intraversion; isc: Intersystem crossing; VR: Vibratory relaxation.

Fluorescence imaging

Fluorescence spectrum imaging technology is a new analysis and detection technology that combines space imaging technology and spectral analysis technology. It processes and analyzes color, texture, fluorescence intensity and other information from the fluorescence images of the test sample. The technology has the advantages of spectral resolution and has spatial resolution to reflect the overall information of the test sample. This method has the advantages of requiring a small sample amount, nondestructive, fast and visual. The technology can be implemented in a hyperspectral or multispectral mode.

Davis et al. (17) used fluorescence hyperspectral imaging for the label-free measurement of algal triacylglyceride production. In their study, they employed hyperspectral fluorescence microscopy with multivariate analysis to track the morphological dynamics and local concentrations of chlorophyll, carotenoids, and lipids in multiple microalgal species, including *Nannochloropsis* sp., *Neochloris oleoabundans*, *Dunaliella salina* and *C. reinhardtii*. By quantifying the volume of carotenoid-containing globules outside the chloroplast, they determined the absolute amount of stored lipid on a per cell basis and compared these results to analysis using traditional Nile Red staining. They proved that this imaging strategy should be generally applicable to novel and recently isolated strains of interest for algal biofuel development and is amenable for sampling assays from large-scale algae biomass production systems.

Moudrikova et al. (110) used confocal Raman spectrograph to get chlorophyll fluorescence from algae cells stained with Nile red. In the experiment, a green 532-nm laser was used to extract chlorophyll fluorescence at low power and with a short time exposure. Then, after restraining chlorophyll fluorescence, they used the 532-nm laser for

Table 6. Applications of fluorescence spectroscopy for determination of microalgae life information.

Species	Analyte	Spectral range (nm)	Method	Evaluation parameters	Advantages	Limitations	Reference
<i>Nannochloropsis</i> <i>Dunaliella</i> <i>Chlamydomonas</i> <i>Neochloris</i>	Triacylglyceride	500–800 nm	–	–	In vivo, labelfree imaging,	The spectral signal is easily interfered	(17)
<i>Microcystis aeruginosa</i>	Microalgae proteins	300–900 nm	–	–	Higher sensitivity and selectivity in microalgae protein detection	The lack of protection of adsorbed groups makes GNPs susceptible to aggregation.	(107)
<i>Porphyridium cruentum</i>	Chlorophyll a	440 nm	–	–	Explained the survival mechanism of red algae in nitrogen-deficient environments.	Needs sample dilution	(108)
6 species of microalgae	Proteins Microalgae density	200–550 nm 260–650 nm	–	–	Analysis of fluorescently dissolved organics produced by different microalgae.	Lack of predictive models	(109)
<i>Chlorella vulgaris</i>	–	–	SVD	–	In a single cell, simultaneous detection	Lack of a more powerful method of spectral processing	(110)
<i>Blue-green algae</i>	Phycocyanin	350 ~ 700 nm	IMC-PLS	–			(111)
<i>Green algae</i>	Chlorophyll a	560–750 nm					
	Chlorophyll a	350–700 nm	IMC-PLS	–	More types of recognition, higher recognition rate	No real-time monitoring is possible	(111)
<i>Diatom</i>	Chlorophyll b	560–750 nm				Selecting wavelengths is time consuming	(111)
	Chlorophyll a	350–700 nm	IMC-PLS	–			
	Chlorophyll C	560–750 nm					
<i>Cryptophyta</i>	Carotenoid						(111)
	Chlorophyll a	350–700 nm	IMC-PLS	–			
	Phycocyanin	560–750 nm					
<i>Cyanobacteria,</i> <i>Green algae,</i> <i>Diatom, Diatom</i>	Chlorophyll C						
<i>Cryptalgae</i>	Carotenoid						
<i>Dunaliella salina</i>	Chlorophyll a	350–700 nm 560–750 nm	MLR	–	Information rich, A variety of algae classification measurements	Only suitable for testing and analysis under laboratory conditions	(112)
	Cell concentration						
	Viability						
		250–690 nm 250–690 nm	PLS-PCA PLS-PCA	$R^2=0.8$ $R^2=0.9$	Real-time monitoring, noninvasive	The quality of the mathematical correlations was not high	(113) (113)
<i>Dunaliella salina</i> and <i>Tetraselmis Chui</i>	Chlorophyll fluorescence	680 nm	–	–	Monitored quantitatively	Chemical inactivation is needed	(114)
<i>Chlorella vulgaris</i>	Chlorophyll fluorescence	564–606 nm	–	–	Detect the changes of cells on a cellular level	No explanation of the fluorescence spectrum	(115)

another scan to measure non-fluorescent biomolecules, such as lipids, proteins, starch and polyphosphate. The results show that fluorescence technology is a powerful tool for characterizing chlorophyll and carotenoid in algae cells and their main energy storage structure.

Sá et al. (113) developed a noninvasive monitoring method that can monitor cell concentration and integrity in real time using two-dimensional (2D) fluorescence spectroscopy. They used PCA and PLS methods to extract information from the fluorescence spectrum and developed models that can predict the cell concentration and survival based only on the fluorescence signal. The R^2 for each model was 0.8 and 0.9, respectively. The article demonstrates that 2D fluorescence spectroscopy has the ability to monitor the cell concentration and integrity of *D. salina* as well as the potential for monitoring algal cell damage during cell harvest.

The application of fluorescence imaging for the monitoring of life information of microalgae is the progress of fluorescence spectroscopy in the field of microalgae. The above results show that fluorescence imaging technology obtains more information than fluorescence spectroscopy and shows great potential in the detection of microalgal life information. This technology can simulate human visual function, extract information such as color, texture, and fluorescence intensity from the fluorescence image of algae for processing and analysis, and finally be used for actual detection. It is an important method for the detection of microalgal life information. However, there is a lack of research on spectral image processing.

Chlorophyll fluorescence

The fluorescence emitted by plants is mainly composed of blue-green fluorescence, red fluorescence and far-red fluorescence. Blue-green fluorescence, also known as short wave fluorescence, is related to the styrene acid in plant cells. Red fluorescence is related to chlorophyll in plant photosystem II. Far-red fluorescence is related to chlorophyll in photosystem I and photosystem II. Therefore, red fluorescence and far-red fluorescence are also known as chlorophyll fluorescence.

Chlorophyll fluorescence has been widely studied and applied as a probe for photosynthesis. Chlorophyll fluorescence can reflect the original reaction process of photosynthesis, such as light energy absorption, excitation energy transfer and photochemical reaction. Almost all changes in the photosynthesis process can be reflected by studying chlorophyll fluorescence. Therefore, chlorophyll fluorescence is an effective tool for detecting life information of microalgae. In recent years, chlorophyll fluorescence was used to monitor the life information of microalgae. Wang et al. (114) used *D. salina* and *Tetraselmis chui* as experimental samples to study the effect of pH on cell capture. A new method for capturing and monitoring single microalgae cells in microfluidic chips is proposed. The survival state of individual microalgae can be quantitatively monitored by measuring the intensity of chlorophyll fluorescence in microalgae. Jiao et al. (115) also used chlorophyll fluorescence to monitor the activity of microalgal cells to study the effects of phosphorus stress on photosynthesis and physiological characteristics of *C. vulgaris*.

Zhao et al.(108) used red algae as a research sample and used a multi-color pulse amplitude modulation (PAM) chlorophyll fluorometer to study the fluorescence kinetics of Chl under nitrogen deficiency conditions to obtain the photosynthetic characteristics of red algae. The results of the study showed that the photochemical efficiency of PSII and the activity of the PSII reaction center (RCs) were reduced because of nitrogen starvation. Additionally, the ability of nitrogen starved cells to withstand and resist high photon flux densities was diminished. The results of their research show that the use of fluorescence kinetics can demonstrate the survival mechanisms of red algae in a nitrogen-deficient environment.

The above results show that chlorophyll fluorescence can be used as photosynthetic probes to monitor the growth of microalgae cells under stress conditions. However, there is a lack of suitable spectral processing algorithms in the above literatures.

3D fluorescence. With the development of three-dimensional (3D) fluorescence spectrometry, the application of 3D fluorescence spectroscopy in phytoplankton research has been gradually expanded, and the fingerprint information provided by this technique is more abundant.

Baowei Ren et al. (109) used 3D fluorescence spectroscopy to monitor the fluorescent dissolved organic matter produced by *Skeletonema costatum*, *Alexandrium tamarense*, *Alexandrium minimatum*, *Scrippsiella trochoidea*, *Prorocentrum donghaiense* and *Prorocentrum micans* in laboratory cultures. The excitation wavelengths and emission wavelengths were set to 200–550 nm and 250–650 nm, respectively. The results showed that the microalgae showed different fluorescence characteristics in different growth stages. According to the change of fluorescence intensity of the proteins, changes in the density of microalgae could be determined. The density of algae could be distinguished in the actual red tide monitoring.

However, the structure of the 3D fluorescence spectrometer is sophisticated and complex. The spectral analysis algorithm is computationally intensive and is currently only suitable for testing and analysis under laboratory conditions. Therefore, it does not meet the needs for the rapid measurement of phytoplankton in the field.

Other applications

Because of the low fluorescence intensity of microalgae, Liu et al. (107) combined microalgae protein with Gold Nanorods (GNRs), and the polymer produced a strong fluorescence signal. Thus, a new method of microalgae protein analysis has been developed. This method is called the method of resonance light scattering system. The best concentration conditions of pH and inorganic salt ions Na^+ were also determined. The results showed that the fluorescence method can quantitatively detect the concentration of microalgal protein, and it can be used as an effective method for biological diagnosis.

To solve the problem of redundancies in the fluorescence spectrum analysis of algae fluorescence, based on the PLS method, Yu et al. (111) proposed an interval Monte Carlo partial least squares (IMC-PLS) method. This method was proposed to effectively solve the problem of selecting the characteristic wavelength. They used the actual measured fluorescence spectra of four kinds of algae to test the proposed method and

compared the data with the commonly used uninformative variable elimination (UVE) to analyze the effectiveness of the method. In the experiment, F-7000 fluorescence spectrophotometer (Hitachi High-Technologies, Tokyo, Japan) was used for measurement. The scanning range was: excitation wavelength of 350–700 nm and emission wavelength of 560–750 nm. The experimental results confirmed the superiority of the proposed method.

Yin et al. (112) analyzed the 3D fluorescence spectra of five species of phytoplankton, including cyanobacteria, green algae, diatoms, dinoflagellates and cryptophytes. The authors extracted 36 characteristic fluorescent spectral points associated with light and pigment. Thus, a method for classifying and measuring different phytoplankton species based on the characteristic fluorescent spectra of pigments was proposed. The experimental results show that this method yielded excellent measurement results for five kinds of algal purebred samples, but the amount of data and calculation time were only 1 and 2.2% of the 3D fluorescence spectroscopy, which proves that this method is a fast and effective method for the classification of algae.

These results indicate that fluorescence technology, especially fluorescence imaging, is a sensitive and effective tool for detecting microalgae life information. The selection of appropriate modeling methods and spectral preprocessing methods play an important role in improving the prediction performance.

Conclusions and future outlook

We attempted to review the recent applications of spectroscopy in algal research, particularly its use for the determination of microalgae life information, such as lipid content, pigment content, protein content, and biomass content as well as the identification of algae species.

The applications of some nondestructive and noninvasive spectroscopic techniques, including VIS and NIR spectroscopy, FTIR spectroscopy, Raman spectroscopy and fluorescence spectroscopy in tandem with chemometrics, such as PCA, PLSR, MLR, LS-SVM and ANN, have been proven for their great potential for the detection of microalgae life information, and satisfactory results have been achieved in related research.

However, most of the research data we found are off-line non-real-time detection of microalgae. The realization of real-time and online detection of microalgae life information is a new challenge for the application of spectrum technology to the microalgae industry.

Due to the inevitable deficiencies in the accuracy and robustness of the information detection models established using chemometric methods, the models lack stability, and the speed of their on-line detection is not very satisfactory. Therefore, in future work, we should strengthen the development of new algorithms for the purpose of improving the accuracy and sensitivity of the model. This includes the extraction of spectral data, the processing of spectral images and the establishment of models. The algorithm must be simple, high-performing and efficient. Under the premise of high accuracy, we should improve the detection speed of microalgae information, develop methods for realizing on-line monitoring of microalgae vital information, and achieve remote data manipulation and sharing. When studying the changes in the spectral and image information during the growth of microalgae, the changes in life information of microalgae at various stages can be clearly revealed.

Acknowledgments

The authors thank American Journal Experts (AJE) for providing English language editing of this paper.

Funding

The authors are grateful for financial support from the Top-notch talent plan program of higher education in Hebei [BJ2017036].

References

1. Wu, D., and Sun, D.-W. (2013) Advanced applications of hyperspectral imaging technology for food quality and safety analysis and assessment: A review-Part II: Applications. *Innov. Food Sci. Emerg.* 19: 15–28.
2. Moreno, G. I. (2008) Microalgae immobilization: Current techniques and uses. *Bioresource Technol.* 99 (10): 3949–3964.
3. Khan, M.-I., Shin, J.-H., and Kim, J.-D. (2018) The promising future of microalgae: Current status, challenges, and optimization of a sustainable and renewable industry for biofuels, feed, and other products. *Microb. Cell Fact.* 17(1): 36.
4. Brennan, L., and Owende P. (2010) Biofuels from microalgae-a review of technologies for production, processing, and extractions of biofuels and co-products. *Renew Sustain Energy Rev.* 14: 557–577.
5. Spitz, T. T., Bergman, M., Moppes, V. D., Grossman, S., and Arad, S. (2005) Antioxidant activity of the polysaccharide of the red microalga *Porphyridium* sp. *J. Appl. Phycol.* 17(17): 215–222.
6. Cerón-García, M.-C., González-López, C.-V., Camacho-Rodríguez, J., López-Rosales, L., García-Camacho, F., and Molina-Grima, E. (2018) Maximizing carotenoid extraction from microalgae used as food additives and determined by liquid chromatography (HPLC). *Food Chem.* 257: 316–324.
7. Xu, Y., Lu, A., Chen, X., Su, D., Wang, W., and Zhang, Y. (2018) Isolation, identification and growth optimisation of freshwater microalgae. *Turk. J. Fish. Aquat. Sci.* 18(11): 1293–1302.
8. Wu, D. (2011) *Research on rapid and non-invasive determination techniques for microalgal growth information and qualities of algal products*. Doctoral dissertation, Zhejiang University, Hangzhou. Retrieved from www.cnki.net.
9. Eswaran, K., Mairh, O.-P., and Rao, P. (2002) Inhibition of pigments and phycocolloid in a marine red alga *Gracilaria edulis* by ultraviolet-B radiation. *Biol. Plantarum.* 45(1): 157–159.
10. Harms P., Kostov Y., and Rao G. (2002) Bioprocess monitoring. *Curr. Opin. Biotechnol.* 13(2):124–127.
11. Simmons, L.-J., Sandgren, C.-D., and Berges, J.-A. (2016) Problems and pitfalls in using HPLC pigment analysis to distinguish Lake Michigan phytoplankton taxa. *J. Great Lakes Res.* 42(2): 397–404.
12. Sá, C., Leal, M.-C., Silva, A., Nordez, S., André, E., Paula, J., and Brotas, V. (2013) Variation of phytoplankton assemblages along the Mozambique coast as revealed by HPLC and microscopy. *J. Sea Res.* 79: 1–11.
13. Peniuk, G.-T., Schnurr, P.-J., and Allen, D.-G. (2016) Identification and quantification of suspended algae and bacteria populations using flow cytometry: Applications for algae biofuel and biochemical growth systems. *J. Appl. Phycol.* 28(1): 95–104.
14. Eckford-Soper, L. K., Davidson, K., and Bresnan, E. (2013) Identification and quantification of toxic and non-toxic strains of the harmful dinoflagellate *Alexandrium tamarense*

- using fluorescence in situ hybridization and flow cytometry. *Limnol. Oceanogr.* 11: 540–548.
15. Wu, D., and Sun, D.-W. (2013) Colour measurements by computer vision for food quality control – A review. *Trends Food Sci. Tech.* 29(1): 5–20.
 16. Jackman, P., Sun, D., Du, C., and Allen, P. (2009) Prediction of beef eating qualities from colour, marbling and wavelet surface texture features using homogenous carcass treatment. *Pattern Recogn.* 42(5): 751–763.
 17. Davis, R.-W., Jones, H.-D.-T., Collins, A.-M., Ricken, J.-B., Sinclair, M.-B., Timlin, J.-A., and Singh, S. (2014) Label-free measurement of algal triacylglyceride production using fluorescence hyperspectral imaging. *Algal Res.* 5: 181–189.
 18. Chen, Y., Sun, D., Cheng, J., and Gao, W. (2016) Recent advances for rapid identification of chemical information of muscle foods by hyperspectral imaging analysis. *Food Eng. Rev.* 8(3): 336–350.
 19. Lu, Y., and Lu, R. (2017) Non-destructive defect detection of apples by spectroscopic and imaging technologies: A review. *T. ASABE.* 60(5): 1765–1790.
 20. Cheng, J., and Sun, D. (2015) Recent applications of spectroscopic and hyperspectral imaging techniques with chemometric analysis for rapid inspection of microbial spoilage in muscle foods. *Compr. Rev. Food Sci. F.* 14(4): 478–490.
 21. Lee, T., Chang, J., and Wang, H. (2013) Current developments in high-throughput analysis for microalgae cellular contents. *Biotechnol. J.*, 8(11SI): 1301–1314.
 22. Murdock, J. N., and Wetzel, D. L. (2009) FT-IR Microspectroscopy enhances biological and ecological analysis of algae. *Appl. Spectrosc. Rev.* 44(PII 9113962684): 335–361.
 23. Podevin, M., Fotidis, I. A., and Angelidaki, I. (2018) Microalgal process-monitoring based on high-selectivity spectroscopy tools: status and future perspectives. *Crit. Rev. Biotechnol.* 38(5): 704–718.
 24. Havlik, I., Lindner, P., Scheper, T., and Reardon, K. F. (2013) On-line monitoring of large cultivations of microalgae and cyanobacteria. *Trends Biotechnol.* 31(7): 406–414.
 25. Volkan Bilgili, A., van Es, H.-M., Akbas, F., Durak, A., and Hively, W.-D. (2010) Visible-near infrared reflectance spectroscopy for assessment of soil properties in a semi-arid area of Turkey. *J. Arid Environ.* 74(2): 229–238.
 26. Curcio, D., Ciraolo, G., D Asaro, F., and Minacapilli, M. (2013) Prediction of soil texture distributions using VNIR-SWIR reflectance spectroscopy. *Procedia Environ. Sci.* 19: 494–503.
 27. Cortés, V., Cubero, S., Aleixos, N., Blasco, J., and Talens, P. (2017) Sweet and nonsweet taste discrimination of nectarines using visible and near-infrared spectroscopy. *Postharvest Biol. Tec.* 133: 113–120.
 28. Jiang, H.-Z., Yoon, S.-C., Zhuang, H., and Wang, W. (2017) Predicting color traits of intact broiler breast fillets using visible and near-infrared spectroscopy. *Food Anal. Methods.* 10(10): 3443–3451.
 29. Melendez-Pastor, I., Almendro-Candel, M., Navarro-Pedreño, J., Gómez, I., Lillo, M., and Hernández, E. (2013) Monitoring urban wastewaters' characteristics by visible and short wave near-infrared spectroscopy. *Water-Sui.* 5(4): 2026–2036.
 30. Song, Y.-X., Ji, J., Mao, C.-Q., Ayoko, G.-A., Frost, R.-L., Yang, Z.-F., and Yuan, X.-Y. (2013) The use of reflectance visible-NIR spectroscopy to predict seasonal change of trace metals in suspended solids of Changjiang River. *Catena.* 109: 217–224.
 31. Khodabakhshian, R., Emadi, B., Khojastehpour, M., and Golzarian, M.-R. (2016) Carob moth, *Ectomyelois ceratoniae*, detection in pomegranate using visible/near infrared spectroscopy. *Comput. Electron. Agr.* 129: 9–14.
 32. Everard, C.-D., McDonnell, K.-P., and Fagan, C.-C. (2012) Prediction of biomass gross calorific values using visible and near infrared spectroscopy. *Biomass Bioenergy.* 45: 203–211.
 33. Guan, X., Chen, X., and Jiang, J. (2011) Integration of modified uninformative variable elimination and successive projections algorithm for determination harvest time of laver by using visible and near infrared spectra. *Afr. J. Arg. Res.* 6(27): 5987–5991.

34. Sakirkin, S. L. P., Morgan, C. L. S., and Auvermann, B. W. (2010) Effects of sample processing on ash content determination in solid cattle manure with visible/near-infrared spectroscopy. *Trans ASABE*, 53(2): 421–428.
35. Ontiveros, R., Diakite, L., Edna Alvarez, M., and Coras, P. (2013) Wastewater salinity assessment using near infrared spectroscopy. *Water Sci. Technol.* 68(4): 879–886.
36. Bovo, B., De Marchi, M., Carlot, M., Soligo, S., Corich, V., and Giacomini, A. (2013) Indirect evaluation of microbial spoiling activity in Grape Marcs by near-infrared spectroscopy. *Am. J. Enol. Vitic.* 64(3): 411–415.
37. Höpfner, T., Bluma, A., Rudolph, G., Lindner, P., and Scheper, T. (2010) A review of non-invasive optical-based image analysis systems for continuous bioprocess monitoring. *Bioprocess Biosyst. Eng.* 33(2): 247–256.
38. Petracci, M., and Baéza, E. (2011) Harmonization of methodologies for the assessment of poultry meat quality features. *Worlds Poult. Sci. J.* 67(01): 137–151.
39. Mulbry, W., Reeves, J., Liu, Y., Ruan, Z., and Liao, W. (2012) Near- and mid-infrared spectroscopic determination of algal composition. *J. Appl. Phycol.* 24(5): 1261–1267.
40. Wei, X., Jiang, L., Zhao, Y., Shao, Y., Qiu, Z., and He, Y. (2016) Study on the nondestructive detection methods for dynamica change of lipid content in *Chlorella* sp. *Spectrosc. Spect. Anal.* 36(5): 1352–1357.
41. Lin, M., Al-Holy, M., Mousavi-Hesary, M., Al-Qadiri, H., Cavinato, A.-G., and Rasco, B.-A. (2004) Rapid and quantitative detection of the microbial spoilage in chicken meat by diffuse reflectance spectroscopy (600–1100 nm). *Lett. Appl. Microbiol.* 39(2): 148–155.
42. Zhou, H., Fang, H., Pan, J., Jiang, L., He, Y., and Shao, Y. (2017) Chlorophyll content research of *Haematococcus pluvialis* based on immersed visible/near-infrared spectroscopy. *Spectrosc. Spect. Anal.* 37(11): 3375–3378.
43. Shao, Y., Pan, J., Zhang, C., Jiang, L., and He, Y. (2015) Detection in situ of carotenoid in microalgae by transmission spectroscopy. *Electron. Agr.* 112(SI): 121–127.
44. Laurens, L. M. L., and Wolfrum, E. J. (2013) High-throughput quantitative biochemical characterization of algal biomass by NIR spectroscopy; multiple linear regression and multivariate linear regression analysis. *J. Agr. Food Chem.* 61(50): 12307–12314.
45. Liu, B., Liu, J., Chen, T., Yang, B., Jiang, Y., Wei, D., and Chen, F. (2015) Rapid characterization of fatty acids in oleaginous microalgae by near-infrared spectroscopy. *Int. J. Mol. Sci.* 16(4): 7045–7056.
46. Ge, Y., and Thomasson, J. A. (2016) NIR reflectance and MIR attenuated total reflectance spectroscopy for characterizing algal biomass composition. *Trans. ASABE*, 59(2): 435–442.
47. Challagulla, V., Walsh, K. B., and Subedi, P. (2014) Biomass and total lipid content assessment of microalgal cultures using near and short wave infrared spectroscopy. *Bioenerg. Res.* 7(1): 306–318.
48. Laurens, L. M. L., and Wolfrum, E. J. (2011) Feasibility of spectroscopic characterization of algal lipids: Chemometric correlation of NIR and FTIR spectra with exogenous lipids in algal biomass. *Bioenerg. Res.* 4(1): 22–35.
49. Brown, M. R., Frampton, D. M. F., Dunstan, G. A., and Blackburn, S. I. (2014) Assessing near-infrared reflectance spectroscopy for the rapid detection of lipid and biomass in microalgae cultures. *J. Appl. Phycol.* 26(1): 191–198.
50. Samek, O., Zemánek, P., Jonáš, A., and Telle, H.-H. (2011) Characterization of oil-producing microalgae using Raman spectroscopy. *Laser Phys. Lett.* 8(10): 701–709.
51. Huang, Y. Y., Beal, C. M., Cai, W. W., Ruoff, R. S., and Terentjev, E. M. (2009) Micro-Raman spectroscopy of algae: Composition analysis and fluorescence background behavior. *Biotech. Bioeng.*, 105(5), 889–898.
52. Oh, S., Yoo, S.-J., Jeong, D.-H., and Lee, J.-M. (2013) Real-time estimation of glucose concentration in algae cultivation system using Raman spectroscopy. *Bioresource Technol.* 142: 131–137.
53. Mats, J., Alexandra, W., and Katarina, A. (2015) Imaging and identification of marine algal bioactive compounds by surface enhanced Raman spectroscopy (SERS). *Natural Prod. Marine Algae.* 1308: 365–374.

54. Jaeger, D., Pilger, C., Hachmeister, H., Oberländer, E., Wördenweber, R., Wichmann, J., Mussgnug, J.-H., Huser, T., and Kruse, O. (2016) Label-free in vivo analysis of intracellular lipid droplets in the oleaginous microalga *Monoraphidium neglectum* by coherent Raman scattering microscopy. *Sci. Rep.* 6(1): 35340.
55. Qin, J., Chao, K., and Kim, M. (2016) Raman scattering for food quality and safety assessment. In *Light Scattering Technology for Food Property, Quality and Safety Assessment*. Lu R., Ed., CRC Press, Boca Raton.
56. Fang, H., Jiang, L., Pan, J., He, Y., Gong, A., and Shao, Y. (2017) Research on microalgae lipid change under nitrogen-based stress by Raman microspectroscopy. *Spectrosc. Spect. Anal.* 37(10): 3108–3111.
57. Buckley, K., and Ryder, A. G. (2017) Applications of Raman spectroscopy in biopharmaceutical manufacturing: A short review. *Appl. Spectrosc.* 71(6): 1085–1116.
58. Kobayashi, Y., Mayer, S. G., and Park, J. W. (2017) FT-IR and Raman spectroscopies determine structural changes of tilapia fish protein isolate and Surimi under different comminution conditions. *Food Chem.* 226: 156–164.
59. Tahir, H. E., Xiaobo, Z., Zhihua, L., Jiyong, S., Zhai, X., Wang, S., and Mariod, A. A. (2017) Rapid prediction of phenolic compounds and antioxidant activity of Sudanese honey using Raman and Fourier transform infrared (FT-IR) spectroscopy. *Food Chem.* 226: 202–211.
60. Nogales-Bueno, J., Baca-Bocanegra, B., Rooney, A., Hernández-Hierro, J. M., Byrne, H. J., and Heredia, F. J. (2017) Study of phenolic extractability in grape seeds by means of ATR-FTIR and Raman spectroscopy. *Food Chem.* 232: 602–609.
61. Ullah, R., Khan, S., Ali, H., Bilal, M., Saleem, M., Mahmood, A., and Ahmed, M. (2017) Raman-spectroscopy-based differentiation between cow and buffalo milk. *J. Raman Spectrosc.* 48(5): 692–696.
62. Lu, Y.-Z., and Lu, R.-F. (2017) Non-destructive defect detection of apples by spectroscopic and imaging technologies: A review. *Trans. ASABE*, 60(5): 1765–1790.
63. Li, M., Canniffe, D.-P., Jackson, P.-J., Davison, P.-A., FitzGerald, S., Dickman, M.-J., Burgess, J.-G., Hunter, C.-N., and Huang, W.-E. (2012) Rapid resonance Raman microspectroscopy to probe carbon dioxide fixation by single cells in microbial communities. *ISME J.* 6(4): 875–885.
64. Kuzmin, A.-N., Pliss, A., and Kachynski, A.-V. (2013) Biomolecular component analysis of cultured cell nucleoli by Raman microspectrometry. *J. Raman Spectrosc.* 44(2): 198–204.
65. Franzen, L., Mathes, C., Hansen, S., and Windbergs, M. (2013) Advanced chemical imaging and comparison of human and porcine hair follicles for drug delivery by confocal Raman microscopy. *J. Biomed. Opt.* 18(6): 061210.
66. Wang, T., Ji, Y., Wang, Y., Jia, J., Li, J., Huang, S., Han, D., Hu, Q., Huang, W. E., and Xu, J. (2014) Quantitative dynamics of triacylglycerol accumulation in microalgae populations at single-cell resolution revealed by Raman microspectroscopy. *Biotechnol. Biofuels.* 7: 58.
67. Shao, Y.-N., Pan, J., Jiang, L., and He, Y. (2015) Microalgae species identification study with Raman microspectroscopy technology. *Spectrosc. Spect. Anal.* 35(7): 1908–1911.
68. Dementjev, A., and Kostkeviciene, J. (2013) Applying the method of coherent anti-Stokes Raman microscopy for imaging of carotenoids in microalgae and cyanobacteria. *J. Raman Spectrosc.* 44(7): 973–979.
69. He, X.-N., Allen, J., Black, P.-N., Baldacchini, T., Huang, X., Huang, H., Jiang, L., and Lu, Y.-F. (2012) Coherent anti-Stokes Raman scattering and spontaneous Raman spectroscopy and microscopy of microalgae with nitrogen depletion. *Biomed. Opt. Express.* 3(11): 2896–2906.
70. Shao, Y.-N., Fang, H., Zhou, H., Wang, Q., Zhu, Y.-M., and He, Y. (2017) Detection and imaging of lipids of *Scenedesmus obliquus* based on confocal Raman microspectroscopy. *Biotechnol Biofuels.* 10(1): 300.
71. Ji, Y.-T., He, Y.-H., Cui, Y.-B., Wang, T.-T., Wang, Y., Li, Y., Huang, W.-E., and Xu, J. (2014) Raman spectroscopy provides a rapid, non-invasive method for quantitation of starch in live, unicellular microalgae. *Biotechnol. J.*, 9(12): 1512–1518.

72. He, Y.-H., Zhang, P., Huang, S., Wang, T.-T., Ji, Y.-T., and Xu, J. (2017) Label-free, simultaneous quantification of starch, protein and triacylglycerol in single microalgal cells. *Biotechnol. Biofuels*. 10(1): 275.
73. Meksiarun, P., Spegazzini, N., Matsui, H., Nakajima, K., Matsuda, Y., and Sato, H. (2015) In vivo study of lipid accumulation in the microalgae marine Diatom *Thalassiosira pseudonana* using Raman spectroscopy. *Appl. Spectrosc.* 69(1): 45–51.
74. Yonamine, Y., Suzuki, Y., Ito, T., Miura, Y., Goda, K., Ozeki, Y., and Hoshino, Y. (2017) Monitoring photosynthetic activity in microalgal cells by Raman spectroscopy with deuterium oxide as a tracking probe. *Chembiochem*. 18(20): 2063–2068.
75. Weiss, T.-L., Chun, H.-J., Okada, S., Vitha, S., Holzenburg, A., Laane, J., and Devarenne, T.-P. (2010) Raman Spectroscopy analysis of *Botryococcene* hydrocarbons from the green microalga *Botryococcus braunii*. *J. Biol. Chem.* 285(42): 32458–32466.
76. Ishitsuka, K., Koide, M., Yoshida, M., Segawa, H., Leproux, P., Couderc, V., Watanabe, M.-M., and Kano, H. (2017) Identification of intracellular squalene in living algae, *Aurantiochytrium mangrovei* with hyper-spectral coherent anti-Stokes Raman microscopy using a sub-nanosecond supercontinuum laser source. *J. Raman Spectrosc.* 48(1): 8–15.
77. Lee, T. H., Chang, J.-S., and Wang, H.-Y. (2013) Rapid and in vivo quantification of cellular lipids in *Chlorella vulgaris* using near-infrared Raman spectrometry. *Anal. Chem.* 85(4): 2155–2160.
78. Assis, C., Oliveira, L.-S., and Sena, M.-M. (2018) Variable selection applied to the development of a robust method for the quantification of coffee blends using mid infrared spectroscopy. *Food Anal. Method.* 11(2): 578–588.
79. Ropodi, A.-I., Panagou, E.-Z., and Nychas, G.-E. (2018) Rapid detection of frozen-then-thawed minced beef using multispectral imaging and Fourier transform infrared spectroscopy. *Meat Sci.* 135: 142–147.
80. Zhao, H.-R., Wang, X.-Y., Chen, G.-H., and Wen, S. (2004) Identification of wheat varieties by FTIR spectrum. *Spectrosc. Spect. Anal.* 24(11): 1338–1341.
81. Huang, F.-T., Ou, Q.-H., Liu, G., Guo, J., and Sheng, H. (2016) Study of different areas rice of Yunnan by Fourier transform infrared. *Spectrosc. Spect. Anal.* 36(S1): 48–49.
82. Cheng, C.-G., Ruan, Y.-M., and Li, B. (2004) Studied on the identification of *Fructus amomi* from its confusable varieties by Fourier transform infrared spectroscopy. *Spectrosc. Spect. Anal.* 24(11): 1355–1358.
83. Channa, Naseem Aslam, Kalhor, Dost Muhammad, Jahangir, and Taj Muhammad. (2018) Rapid diagnostic method of tobacco products in saliva by Fourier transform infrared spectroscopy (FTIR). *Pak. J. Pharm.* 31(1): 175–180.
84. Eliaerts, J., Dardenne, P., and Meert, N. (2017) Rapid classification and quantification of cocaine in seized powders with ATR-FTIR and chemometrics. *Drug Test Anal.* 9(10): 1480–1489.
85. Gonzalez-Torres, A., Rich, A. M., and Marjo, C. E. (2017) Evaluation of biochemical algal FLOC properties using reflectance Fourier-transform infrared imaging. *Algal Res.* 27(27): 345–355.
86. Kosa, G., Shapaval, V., Kohler, A., and Zimmermann, B. (2017) FTIR spectroscopy as a unified method for simultaneous analysis of intra- and extracellular metabolites in high-throughput screening of microbial bioprocesses. *Microb. Cell Fact.* 16(1): 195.
87. Wang, W.-P., Chai, A., Shi, Y.-X., Xie, X., and Li, B. (2015) Quantitative detection of Chinese cabbage Clubroot based on FTIR spectroscopy. *Spectrosc. Spect. Anal.* 35(05): 1243–1247.
88. Chai, A., Li, B., Shi, Y.-X., and Xie, X. (2011) Early detection of *Corynespora* Spot on cucumber leaves based on FTIR spectroscopy. *Spectrosc. Spect. Anal.* 31(06): 1506–1509.
89. Movasaghi, Z., Rehman, S., and Rehman, I. U. (2008) Fourier transform infrared (FTIR) spectroscopy of biological tissues. *Appl. Spectrosc. Rev.* 43: 134–179.
90. Wagner, H., Dunker, S., Liu, Z., and Wilhelm, C. (2013) Subcommunity FTIR-spectroscopy to determine physiological cell states. *Curr. Opin. Biotechnol.* 24(1): 88–94.

91. Kansiz, M., Heraud, P., Wood, B., Burden, F., Beardall, J., and McNaughton, D. (1999) Fourier transform infrared microspectroscopy and chemometrics as a tool for the discrimination of cyanobacterial strains. *Phytochemistry*. 52(3): 407–417.
92. Rüger, J., Unger, N., Schie, I.-W., Brunner, E., Popp, J., and Krafft, C. (2016) Assessment of growth phases of the diatom *Ditylum brightwellii* by FT-IR and Raman spectroscopy. *Algal Res.* 19: 246–252.
93. Sackett, O., Petrou, K., Reedy, B., Hill, R., Doblin, M., Beardall, J., Ralph, P., and Heraud, P. (2016) Snapshot prediction of carbon productivity, carbon and protein content in a Southern Ocean diatom using FTIR spectroscopy. 10(2): 416.
94. Mayers, J. J., Flynn, K. J., and Shields, R. J. (2013) Rapid determination of bulk microalgal biochemical composition by Fourier-transform infrared spectroscopy. *Bioresource Technol.* 148: 215–220.
95. Dean, A. P., Sigee, D. C., Estrada, B., and Pittman, J. K. (2010) Using FTIR spectroscopy for rapid determination of lipid accumulation in response to nitrogen limitation in freshwater microalgae. *Bioresource Technol.* 101(12): 4499–4507.
96. Horton, R. B., Duranty, E., McConico, M., and Vogt, F. (2011) Fourier transform infrared (FT-IR) spectroscopy and improved principal component regression (PCR) for quantification of solid analytes in microalgae and bacteria. *Appl. Spectrosc.* 65(4): 442–453.
97. Coat, R., Montalescot, V., Leon, E. S., Kucma, D., Perrier, C., Jubeau, S., Thouand, G., Legrand, J., Pruvost, J., and Goncalves, O. (2014) Unravelling the matrix effect of fresh sampled cells for in vivo unbiased FTIR determination of the absolute concentration of total lipid content of microalgae. *Bioproc. Biosyst. Eng.* 37(11): 2175–2187.
98. Wagner, H., Liu, Z., Langner, U., Stehfest, K., and Wilhelm, C. (2010) The use of FTIR spectroscopy to assess quantitative changes in the biochemical composition of microalgae. *J. Biophotonics*. 3(8–9): 557–566.
99. Bounphanmy, S., Thammathaworn, S., Thanee, N., Pirapathrungsuriya, K., Beardall, J., McNaughton, D., and Heraud, P. (2010) Discrimination of cyanobacterial strains isolated from saline soils in Nakhon Ratchasima, Thailand using attenuated total reflectance FTIR spectroscopy. *J. Biophotonics*. 3(8–9): 534–541.
100. Liu, J.-H., and Huang, Q. (2016) Screening of astaxanthin-hyperproducing *Haematococcus pluvialis* using Fourier transform infrared (FT-IR) and Raman microspectroscopy. *Appl. Spectrosc.* 70(10): 1639–1648.
101. Illman, A. M., Scragg, A. H., and Shales, S. W. (2000) Increase in *Chlorella* strains calorific values when grown in low nitrogen medium. *Enzyme Microb. Technol.* 27: 631–635.
102. Pistorius, A. M. A., DeGrip, W. J., and Egorova-Zachernyuk, T. A. (2009) Monitoring of biomass composition from microbiological sources by means of FT-IR spectroscopy. *Biotechnol. Bioeng.* 103:123–129.
103. Gorbe, E., and Calatayud, A. (2012). Applications of chlorophyll fluorescence imaging technique in horticultural research: A review. *Sci Hort.* 138, 24–35.
104. Zhang, R., Ying, Y., Rao, X., and Li, J. (2012). Quality and safety assessment of food and agricultural products by hyperspectral fluorescence imaging. *J. Sci. Food Agric.* 92(12), 2397–2408.
105. C'owles, T. J., Desiderio, R. A., and Neuer, S. (1993) In situ characterization of phytoplankton from vertical profiles of fluorescence emission spectra. *Mar. Biol. Res.* 115(2): 217–222.
106. Lee, T., Tsuzuki, M., Takeuchi, T., Yokoyama, K., and Karube, I. (1995) Quantitative determination of cyanobacteria in mixed phytoplankton assemblages by an in vivo fluorimetric method. *Anal. Chim. Acta.* 302(1): 81–87.
107. Liu, S.-Y., Tang, L., Wang, J.-L., Fu, H.-Z., Wan, X., and Li, Y.-B. (2017) Label-free and sensitive detection of microalgae protein using GNRs-based resonance light scattering system. *RSC Adv.* 7(23): 14039–14045.
108. Zhao, L., Li, K., Wang, Q.-M., Song, X., Su, H., Xie, B., Zhang, X.-Y., Huang, F., Chen, X.-L., Zhou, B.-C., and Zhang, Y.-Z. (2017) Nitrogen starvation impacts the

- photosynthetic performance of *Porphyridium cruentum* as revealed by chlorophyll a fluorescence. *Sci. Rep.* 7(1): 8542.
109. Ren, B.-W., Zhao, W.-H., Wang, J.-T., and Wang, L. (2008) Three dimensional fluorescence excitation-emission matrix spectrum of dissolved organic substance in marine microalgae growth process. *Spectrosc. Spect. Anal.* 28(05): 1130–1134.
 110. Moudrikova, S., Mojzes, P., Zachleder, V., Pfaff, C., Behrendt, D., and Nedbal, L. (2016) Raman and fluorescence microscopy sensing energy-transducing and energy-storing structures in microalgae. *Algal Res.* 16: 224–232.
 111. Yu, X.-Y., Zhang, Y.-J., Yin, G.-F., Xiao, X., Zhao, N., Duan, J., Shi, C., and Fang, L. (2014) Feature wavelength selection of phytoplankton fluorescence spectra based on partial least squares. *Acta. Optic. Sin.* 34(09): 302–307.
 112. Yin, G.-F., Zhao, N.-J., Hu, L., Yu, X., Shi, C., Xiao, X., Fang, L., Duan, J., Gan, T., Zhang, Y., Liu, J., and Liu, W. (2014) Classified measurement of phytoplankton based on characteristic fluorescence of photosynthetic pigments. *Acta. Optic. Sin.* 34(09): 320–325.
 113. Sá, M., Monte, J., Brazinha, C., Galinha, C.-F., and Crespo, J.-G. (2017) 2D Fluorescence spectroscopy for monitoring *Dunaliella salina* concentration and integrity during membrane harvesting. *Algal Res.* 24: 325–332.
 114. Wang, J.-S., Meng, X.-F., Song, Y.-X., Pan, X.-X., and Li, D.-Q. (2016) Detection of activity of single microalgae cells in a new microfluidic cell capturing chip. *Meas. Sci. Technol.* 27(12): 125701.
 115. Jiao, Y., Ouyang, H.-L., Jiang, Y.-J., Kong, X.-Z., He, W., Liu, W.-X., Yang, B., and Xu, F.-L. (2017) Effects of phosphorus stress on the photosynthetic and physiological characteristics of *Chlorella vulgaris* based on chlorophyll fluorescence and flow cytometric analysis. *Ecol. Indic.* 78: 131–141.